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A new era for digital twins: progress and industry adoption

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ABSTRACT

This systematic review provides a comprehensive evaluation of the core components and diverse applications of digital twin (DT) technology, emphasising its transformative influence across key sectors. A DT is a dynamic digital replica of a physical asset or system, developed through the integration of real-time sensor data, advanced communication protocols, and computational intelligence. Essential elements include machine learning for predictive analytics, edge computing for low-latency decision-making, and high-resolution imaging and 3D visualisation for enhanced model fidelity. Blockchain technologies strengthen data security and integrity, while adaptive feedback mechanisms enable continuous learning and system optimisation. DT applications span various industries. In manufacturing, they enhance productivity through predictive maintenance and process refinement. In healthcare, DTs support personalised diagnostics, treatment optimisation, and telemedicine. Urban planning benefits from DTs in the creation of smart, sustainable infrastructure. In the energy sector, they facilitate grid stability and renewable integration. Additionally, DTs are used in immersive training simulations, autonomous vehicle validation, and resilient supply chain management. Industries such as aerospace, automotive, marine, oil and gas, and transportation increasingly adopt DTs to boost operational efficiency, minimise risks, and foster innovation. This review synthesises recent technological advances and identifies critical research directions for advancing secure, scalable, and interoperable DT ecosystems.

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Digital twin (DT); real-time data processing; predictive analytics; cyber-physical systems; smart infrastructure

1. Introduction

Digital twin (DT) technology is a transformative innovation that bridges the physical and digital worlds. It creates virtual replicas of real-world systems, enabling real time monitoring, simulation, and optimisation. By integrating data with computational models, DT enhances efficiency, sustainability, and resilience across various industrial systems, giving operators full visibility of production processes [1].

Originally developed for aerospace applications, DT technology has expanded to offer transformative solutions for complex challenges in numerous sectors, including manufacturing, healthcare, infrastructure and urban planning, energy, logistics, education, transportation, and autonomous systems [2].

DT technology relies on sensors, data acquisition systems, and computational infrastructures for seamless data collection and real time processing. Machine learning and analytics improve predictive capabilities, allowing for proactive decision-making [3]. Connectivity solutions ensure smooth data exchange, while visualisation tools, high-resolution imaging, and scanning techniques enhance digital modelling accuracy [4]. Advanced features in modern IoT devices further improve their performance [5,6]. To ensure security and data integrity, blockchain and cybersecurity mechanisms play a critical role [7].

Digital twins offer several key applications, including predictive maintenance, process optimisation, design and prototyping, and healthcare advancements. In predictive maintenance, digital twins facilitate

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the identification of potential system failures before they occur, thereby reducing downtime, extending equipment lifespan, and lowering operational costs [8]. Process optimisation is another significant application, where digital twins streamline workflows and enhance resource allocation, leading to increased efficiency [7]. Furthermore, design and prototyping benefit from digital twins, as they enable virtual testing, accelerate development cycles, and reduce the need for physical prototypes [9]. In the healthcare sector, digital twins enable patient-specific modelling, which improves diagnosis, treatment planning, and surgical simulations [10].

Digital twins also play a crucial role in smart infrastructure and urban planning. Engineers use them to monitor structural health and predict potential failures [11], while urban planners utilise them to simulate traffic flow and optimise resource distribution, fostering the development of sustainable cities [12]. In the energy sector, digital twins enhance grid operations, facilitate the integration of renewable energy sources, and improve system safety [13]. Furthermore, both renewable and nuclear energy plants employ digital twins for real-time monitoring, efficiency optimisation, and risk assessment [14].

DT also plays a vital role in training and education by providing interactive and immersive simulations that enhance learning and skill development [15]. It improves autonomous systems by creating high-fidelity virtual environments, allowing for safe algorithm testing before deployment [16]. In supply chain management, DT improves demand forecasting, route planning, and real-time tracking, thereby increasing operational efficiency [17]. In manufacturing and aerospace, DT optimises production quality, monitors equipment performance, and facilitates predictive maintenance, leading to improved safety and productivity [18].

The impact of DT extends further into energy production and distribution, where it enhances the reliability of utilities and optimises power management [19]. In civil engineering, DT supports real-time infrastructure monitoring, improving safety and extending the lifespan of critical structures [20]. Robotics and automation also benefit from DT, as it enables real-time system optimisation and process refinement, improving accuracy and efficiency [21]. Oil and gas industries utilise DT to optimise workflows, enhance predictive maintenance, and improve risk management [22]. Similarly, transportation and marine engineering leverage DT for fleet management, route optimisation, and operational safety, reducing costs and improving overall system reliability [23].

Despite its vast potential, DT faces several challenges, including high computational demands, data integration complexity, cybersecurity threats, and a lack of standardisation [24]. To address these challenges requires collaboration among researchers, industry experts, and policymakers to develop scalable and secure DT implementations [7]. Future research should focus on AI-driven analytics, hybrid DT models, edge computing for real-time processing, and blockchain-based security solutions [25]. Establishing standardised frameworks and interdisciplinary training programs will also be essential in driving the widespread adoption of DT across industries.

This systematic review examines the current state of DT technology encompassing commonly integrated elements, its most impactful applications across industries, and future research directions.

This article is structured into 6 comprehensive sections. [Section 1](#) introduces digital twin technology and its diverse applications. [Section 2](#) details the systematic review methodology, including research question formulation, literature search strategies, inclusion and exclusion criteria, study selection procedures, data extraction techniques, and validation methods. [Section 3](#) delves into the core components and underlying technologies of digital twins. [Section 4](#) presents a detailed overview of DT applications across various industries. [Section 5](#) discusses existing technologies, current challenges, and future research directions. Finally, [Section 6](#) concludes the article, summarising key findings and implications.

2. Methodology

This systematic review followed a structured and comprehensive approach to examine the structure and application of various types of digital twins across different industries. The review process was conducted in five distinct stages: defining the research questions, identifying relevant studies, selecting studies based on inclusion and exclusion criteria, extracting data and analysing the findings.

Research Question Formulation: This systematic review aimed to synthesise evidence regarding the applications of different types of digital twins. The review addressed the following specific questions:

- What are the essential components of a functional digital twin system?
- In what industries are digital twins currently being used successfully?

Literature Search Strategy: Searches were conducted using the phrase ‘the elements of a Digital twin’ and ‘the application of Digital twin’ in relation to the following areas: Predictive Maintenance, Process Optimisation, Design and Prototyping, Personalised Medicine, Infrastructure and Urban Planning, Energy Management, Training and Education, Autonomous Systems, Supply Chain and Logistics, Aerospace Engineering, Automotive Engineering, Energy Systems and Utilities, Civil and Structural Engineering, Healthcare Engineering, Industrial Automation and Robotics, Oil and Gas Engineering, Transportation and Logistics, Marine and Offshore Engineering, and Smart Grids and Utilities.

The search was limited to articles published in peer-reviewed journals between 2020 and 2025. To ensure thorough coverage, both backward and forward citation tracking of selected articles was performed.

Inclusion and exclusion criteria: Studies were included/excluded based on the following criteria:

Inclusion criteria:

- Studies published between 2020 to 2025.
- Studies published in English
- Studies that explicitly define or describe the technology being used as a ‘digital twin’.
- Application Area: Studies that describe the application of digital twins in one or more of the following areas:
 - Predictive Maintenance
 - Process Optimisation
 - Design and Prototyping
 - Personalised Medicine
 - Infrastructure and Urban Planning
 - Energy Management
 - Training and Education
 - Autonomous Systems
 - Supply Chain and Logistics
 - Aerospace Engineering
 - Automotive Engineering
 - Energy Systems and Utilities
 - Civil and Structural Engineering
 - Healthcare Engineering
 - Industrial Automation and Robotics
 - Oil and Gas Engineering
 - Transportation and Logistics
 - Marine and Offshore Engineering
 - Smart Grids and Utilities

Exclusion criteria:

- Editorials, letters to the editor, and opinion pieces
- Conferences
- Books and book chapters (unless they contain primary research)
- Patents
- Websites, blogs, and other non-peer-reviewed sources

- Language: Studies published in languages other than English
- Studies published before 2020
- Studies that do not provide a clear definition or description of a digital twin. Studies that use the term 'digital twin' loosely or interchangeably with other concepts (e.g. simulation, modelling) without meeting the core components.
- Application Area: Studies that do not focus on the specified application areas.
- Studies that describe the theoretical concepts of digital twins without empirical data or real-world applications.

Study selection: Following database searches, duplicates were removed. Two independent reviewers then screened the titles and abstracts of the remaining studies and subsequently assessed the full-text articles for eligibility based on the inclusion and exclusion criteria. Disagreements were resolved through discussion.

Data extraction and synthesis: For the selected studies, data were extracted on the following parameters: type of digital twin system used, industrial application, technology used. A qualitative synthesis was performed to analyse the findings and categorise the digital twin applications based on their functionality.

Quality assessment: The quality of the included studies was evaluated using a modified version of the Newcastle-Ottawa Scale, specifically adapted to assess the methodological rigour and relevance of control system used in rehabilitation research.

Data analysis: A total of 464 articles were initially retrieved from multiple databases. Of these, 118 were excluded for lacking relevant information. An additional 20 articles were removed due to duplication. Another 104 articles were excluded because their outcomes did not align with the objectives of this study. After applying these criteria, 222 articles were selected for detailed analysis. [Figure 1](#) presents the flow chart of the search and selection process.

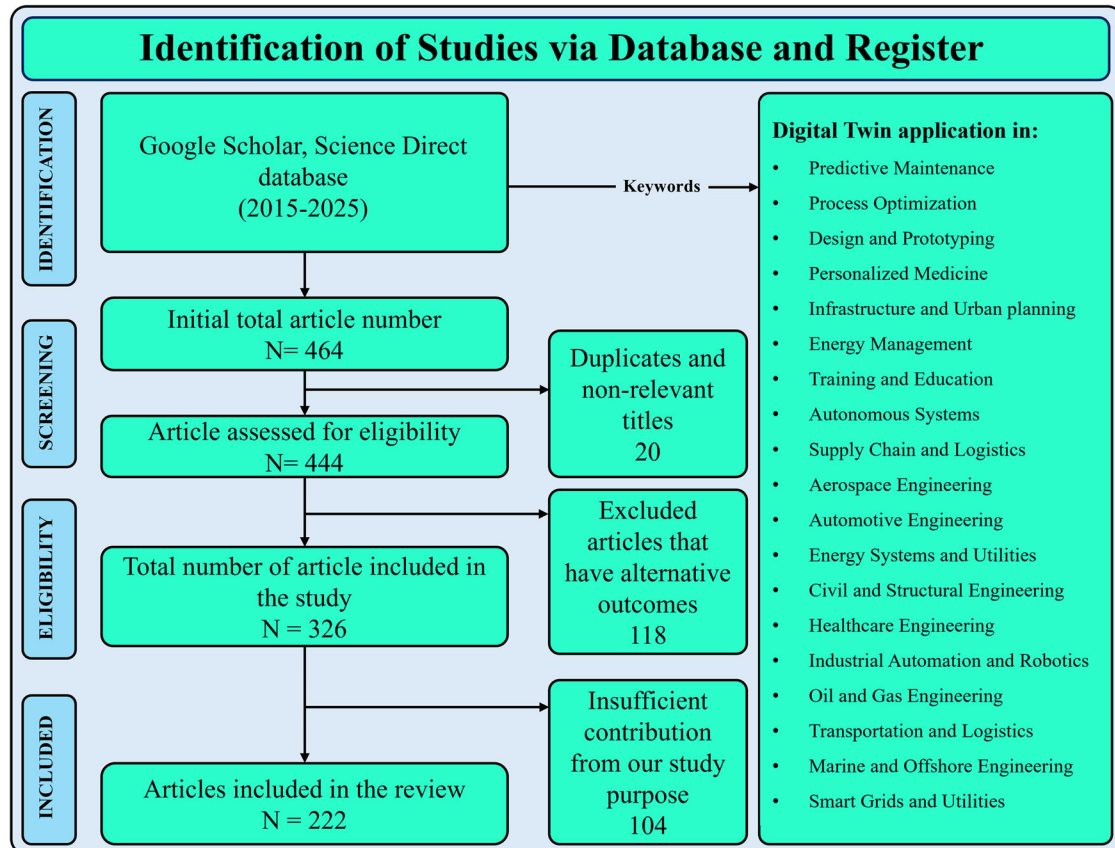


Figure 1. Flow chart of the search and inclusion process.

3. Key elements of digital twin

DT technology creates real-time digital replicas of physical assets, enabling continuous monitoring, simulation, and optimisation. A robust DT system integrates several core components to function effectively. Sensors and IoT devices capture real-time data from the physical environment. Data acquisition and integration systems consolidate information from multiple sources, ensuring accurate and timely inputs.

High-performance computing infrastructure, including cloud platforms, enables the processing and analysis of large-scale data. Simulation and modelling software develop virtual representations of physical systems, while analytics and machine learning tools enhance prediction and support informed decision-making.

Visualisation tools, such as dashboards and 3D simulations, facilitate user interaction by presenting data in an intuitive format. Blockchain technology enhances data security and transparency, while cybersecurity frameworks safeguard the system from evolving digital threats. Edge computing allows for low-latency processing near the data source, improving responsiveness.

Standardised communication protocols ensure smooth interoperability between components. High-resolution imaging technologies like LiDAR and 3D scanners help create precise digital models. Advanced databases support the storage and management of complex, dynamic datasets.

These components together form the foundation of an effective digital twin system. They drive innovation and improve efficiency in industries including manufacturing, healthcare, infrastructure, and smart cities. [Section 3](#) explaining the basic components of the digital twin system.

3.1. Sensors and IoT devices

Sensors and Internet of Things (IoT) devices are essential for DT systems, as they collect real-time data from physical assets to enable accurate monitoring and analysis. Key parameters such as temperature, pressure, vibration, and environmental conditions are captured, providing vital inputs for DT models [24]. Sensors connect the physical and digital realms by transmitting data for processing and simulation. For example, temperature sensors in manufacturing detect heat changes to predict issues [7], while accelerometers identify mechanical anomalies, abnormal vibrations [14]. In healthcare, wearables track vital signs for decision support [10]. Reliable, high-quality sensors minimise errors and ensure dependable system performance [12,15,20].

3.2. Data acquisition and integration systems

Robust data acquisition and integration are critical to the functionality of DT systems. These processes involve aggregating data from various sources such as IoT devices, enterprise systems, and historical datasets, each offering vital inputs for accurately representing physical systems [1]. IoT devices play a key role by capturing real-time environmental and operational data, measuring parameters like temperature, pressure, and motion, which enhances the DT's ability to mirror its physical counterpart and deliver real-time performance insights [1,24].

Enterprise systems, including Enterprise Resource Planning (ERP), Customer Relationship Management (CRM), and supply chain platforms, provide essential operational and transactional data. Integrating this information into DT systems enables a holistic understanding of organisational operations, supporting improved efficiency and decision-making [26,27]. Historical datasets contribute by revealing trends, patterns, and past performance, which strengthen predictive analytics and support forecasting when combined with real-time data [1,6].

Advanced integration tools are vital for harmonising these diverse sources. They facilitate real-time communication, reduce delays, and standardise data formats for interoperability [27,28]. This seamless integration is crucial for timely anomaly detection, failure prediction, and operational optimisation—especially in sectors like manufacturing, healthcare, and logistics [29]. Ultimately, effective data acquisition and integration empower organisations with real-time operational visibility and smarter decision-making, driven by unified, high-quality data inputs [30].

3.3. Computational infrastructure

High-performance computing (HPC) infrastructure is essential for managing the vast data volumes generated by DT systems. These systems require significant computational power for real-time simulation, analysis, and optimisation of complex processes [31,32]. Cloud computing platforms, such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud, offer scalable solutions to meet these demands [33,34]. Their flexibility allows organisations to increase computing capacity as needed without major investments in physical infrastructure [35,36]. This scalability supports dynamic workloads and evolving operational needs. Beyond data storage and processing, cloud platforms also provide advanced tools, including machine learning, analytics engines, and data visualisation services [37]. These capabilities enable businesses to gain actionable insights, optimise performance, and enhance decision-making in DT systems [38].

3.4. The role of simulation and modelling software in Digital Twin systems

Simulation and modelling software are fundamental to DT systems, providing the tools needed to create virtual replicas of physical assets for real-time simulations, performance monitoring, and detailed analysis [4]. By accurately mirroring real-world systems, these tools support optimisation, cost reduction, and improved decision-making [24], with their integration being vital to generating reliable and actionable insights [32].

Popular platforms include ANSYS, Siemens NX, and MATLAB, each tailored for specific applications [39]. ANSYS offers robust tools for structural, thermal, and fluid simulations [34]. Siemens NX is widely used in automotive and aerospace industries for integrated design and manufacturing [4]. MATLAB is valuable in research and development, excelling in numerical computing and system modelling.

A key strength of these platforms lies in combining physics-based modelling with data-driven methods [24]. Physics-based models simulate system behaviours using scientific principles such as stress analysis or fluid dynamics [15,39]. However, real-world complexity often requires machine learning and AI to detect patterns and anomalies that traditional models might miss [6,9].

Real-time simulation, enabled through sensor and IoT data integration, supports continuous asset monitoring and proactive decision-making [1,24,40]. These capabilities help detect equipment wear early, optimise maintenance, and minimise downtime [32].

Additionally, simulation tools accelerate product development by enabling virtual prototyping. Engineers can explore materials, configurations, and designs without physical prototypes, reducing cost and time while encouraging innovation [6,15,32].

3.5. Analytics and machine learning (ML) tools

Advanced analytics and machine learning (ML) tools play a vital role in unlocking the full potential of DT models. These computational tools convert raw, multi-dimensional data into meaningful insights through pattern recognition, anomaly detection, and predictive modelling techniques [9]. ML algorithms are particularly effective at identifying hidden trends within large datasets, enabling early fault detection and accurate forecasting of future states based on real-time and historical data streams [24].

One of the most significant applications of ML in DT environments is predictive maintenance. By analysing simulation outputs and sensor data from physical assets, ML models can detect degradation patterns and predict component failures before they occur. This enables condition-based maintenance scheduling, thereby reducing unplanned downtimes, extending asset lifespans, and lowering maintenance costs [15,40]. In addition, anomaly detection through ML algorithms helps identify abnormal behaviours in operational data, allowing organisations to proactively address safety risks and inefficiencies [41].

ML tools also support process optimisation by analysing DT data to uncover inefficiencies in workflows, resource allocation, and system configurations. They provide data-driven recommendations to

streamline operations, improve productivity, and minimise waste, especially important in industries where precision and efficiency are critical [4,37].

Furthermore, real-time ML-driven analytics offer a dynamic view of current performance, supporting adaptive decision-making and enhanced resource utilisation [42]. These tools are foundational for deriving predictive and prescriptive intelligence from DTs. By leveraging ML, organisations can achieve higher reliability, responsiveness, and operational excellence in increasingly complex and data-rich environments [27].

3.6. Visualisation tools

Visualisation tools are crucial for enabling intuitive interaction with DT models, bridging complex datasets and actionable insights [23]. By presenting data through user-friendly interfaces, these tools facilitate clearer interpretation and more informed decision-making [43,44]. Core functionalities include dashboards, which aggregate multiple data streams into a cohesive display for real-time monitoring, anomaly detection, and performance evaluation [32,45,46].

3D simulations are another critical component, offering virtual replicas of physical systems. These allow for in-depth visualisation, diagnostics, and experimentation without physical risk [9,15, 47,48]. Augmented reality (AR) interfaces further enhance user interaction by overlaying digital data on the real world, improving task accuracy and operational efficiency [1,43,49].

Platforms like Unity and Unreal Engine provide immersive 3D environments widely adopted in sectors such as manufacturing and automotive [39,45,50]. Meanwhile, specialised AR platforms integrated with IoT devices deliver real-time, interactive insights for enhanced situational awareness [1,32,49].

Applications span multiple domains-monitoring and optimising production in manufacturing, advancing training in healthcare, and improving planning in infrastructure management [47–49].

In summary, visualisation tools-through dashboards, simulations, and AR-translate data into actionable knowledge [45]. With platforms like Unity and Unreal Engine, these tools empower effective DT interaction, enhancing operations and innovation across industries [9,49,51].

3.7. Blockchain technology

Blockchain enhances data security and transparency in DT systems by creating immutable transaction records, ensuring data integrity [52–54].

Blockchain secures sensitive data in sectors like healthcare and supply chain management, preventing unauthorised access [54–57]. It also provides transparency through a decentralised ledger, allowing stakeholders to trace data history [54,58]. This is valuable in collaborative environments like smart energy grids [59].

By providing a single source of truth and verifying transactions through consensus mechanisms, blockchain enhances trust and accountability [54,57,60]. Its immutability creates a robust audit trail, useful in supply chains for resolving disputes [60]. Smart contracts can automate processes, such as payment release upon delivery verification, streamlining operations [57,61,62].

Blockchain improves data security, transparency, trust, and accountability in DT systems. Its immutable records and decentralised nature support secure, efficient, and trustworthy operations across diverse industries [53,54,61].

3.8. Cybersecurity frameworks

As connectivity grows, cybersecurity is a vital concern for DT systems, which are vulnerable to cyber threats due to constant data exchange, [44,62]. Robust cybersecurity measures are essential to ensure their reliability and safety [45]. A comprehensive security framework is crucial for protecting DT systems and ensuring data confidentiality and integrity [63–65].

Key components include encryption, which secures data during transmission and storage [44,66,67] and access control, which defines and manages user access through methods like role-based access

controls and multi-factor authentication [62,63,68]. Real-time threat detection using machine learning and AI is also essential for identifying and responding to potential threats [64–68].

The importance of cybersecurity spans industries, including manufacturing, healthcare, and energy, where cyberattacks could have severe consequences [63,64,68].

Implementing strong cybersecurity measures also builds trust among stakeholders, which is essential for the adoption and growth of DT technologies [62,65].

In conclusion, cybersecurity is critical for DT systems. Robust frameworks with encryption, access control, and real-time threat detection are necessary to protect data, ensure reliable operations, and foster trust. The importance of strong security measures will only increase as cyber threats evolve [44,63].

3.9. Edge computing devices

Edge computing devices process data locally, reducing reliance on centralised cloud infrastructure and improving response times for real-time applications [69,70].

DT systems with real-time processing needs benefit from edge computing, which processes data closer to its source (sensors, IoT devices) instead of relying on a centralised cloud [69,71]. This reduces latency, which is crucial for applications like autonomous vehicles, industrial automation, and health-care monitoring [9,69,71]. Edge computing also alleviates network strain and reduces data transfer, enhancing efficiency and lowering costs [69]. Edge solutions can operate in areas with limited connectivity [69].

Cloud computing handles large-scale data storage and long-term trend monitoring, while edge computing provides immediate processing and low-latency responses [69,70]. This combination allows organisations to maximise their DT systems' potential [9].

3.10. Digital communication protocols

Standardised communication protocols are fundamental to achieving effective interoperability in DT systems [72]. These protocols enable seamless data exchange between physical assets, IoT devices, and digital models across diverse platforms, enhancing both the scalability and flexibility of DT implementations [73]. They serve as a common language for data exchange [74], which is critical given that DT ecosystems often incorporate components from multiple manufacturers and platforms [75]. In the absence of standardisation, communication between heterogeneous systems can be inefficient or even fail, resulting in compatibility issues and operational disruptions [76]. To bridge these gaps, protocols such as MQTT, OPC UA, and REST APIs are commonly employed in DT architectures [32].

MQTT (Message Queuing Telemetry Transport) is favoured for its lightweight architecture and efficiency [77]. It is ideal for IoT-based DT systems operating under bandwidth or resource constraints [78]. Operating on a publish-subscribe model, MQTT minimises latency, making it particularly suitable for real-time applications like equipment monitoring or environmental tracking [32,79].

OPC UA (Open Platform Communications Unified Architecture) provides a robust and versatile protocol for industrial DT systems. Its platform independence allows seamless integration across hardware and software environments issues [76,80]. Moreover, it includes built-in security features such as encryption and authentication making it appropriate for use in security-sensitive sectors [81].

REST APIs (Representational State Transfer Application Programming Interfaces) offer another powerful tool for interoperability. Using standard HTTP methods, REST APIs enable efficient communication between DT modules, devices, and external applications [82,83]. Their simplicity and widespread adoption make them especially effective for integrating DT systems with cloud services and third-party platforms [32].

In addition to facilitating real-time communication, standardised protocols enhance the scalability of DT ecosystems by allowing new components to be added without major reconfigurations [80,82]. They also support flexibility, enabling DT systems to adapt to evolving technologies and dynamic business requirements [83–85].

3.11. High-Resolution imaging and scanning equipment

Technologies such as LiDAR, 3D scanners, and high-resolution cameras are essential for developing accurate virtual models in DT systems [85–87]. These tools capture the physical features of objects and environments, enabling detailed and reliable digital representations. LiDAR uses laser beams to measure distances and create clear 3D models, even under varying lighting conditions [87–90]. 3D scanners, employing laser or structured light, are widely used in manufacturing for reverse engineering and quality control [85–92]. High-resolution cameras enhance virtual realism by capturing textures and colours, supporting applications like VR simulations [85]. In infrastructure, LiDAR and 3D scanning support project visualisation and design optimisation, leading to cost and time savings [93,94]. In healthcare, 3D scanning improves diagnostics and treatment planning while other industries benefit from immersive and innovative digital experiences [94–96].

3.12. Advanced databases

Advanced database systems are vital for managing the vast and complex datasets required by DT systems, ensuring scalability, performance, and reliability [97]. Distributed databases like Cassandra and MongoDB are widely used in DT implementations to handle large volumes of data across multiple servers, supporting fault tolerance and high availability [98–102].

Cassandra excels in write-heavy, IoT-based DT systems [98,99], while MongoDB's flexible schema enables efficient mapping of diverse datasets to virtual models [86,103–105]. Time-series databases, such as InfluxDB, are essential for managing real-time, time-stamped data from sensors in dynamic environments [97,106,107]. Integrating distributed and time-series databases allows for hybrid storage architectures that enhance system performance [97,104,106]. These database systems also support analytics and visualisation tools, improving decision-making, operational efficiency, and system reliability across industries [99,103, 105–109].

4. Comprehensive applications of DT

DT technology is transforming industries by enabling real-time monitoring, simulation, and predictive analytics through virtual models of physical assets. Key applications include predictive maintenance to reduce downtime and process optimisation in manufacturing, healthcare, and logistics. In design and prototyping, DTs allow early testing, reducing costs and improving product quality. Healthcare uses DTs for personalised treatment planning, while urban planning benefits from improved traffic flow and infrastructure design. In energy, DTs enhance grid reliability and sustainability. They also support risk-free training in healthcare and aviation. Autonomous systems and robotics use DTs for learning and operations. DTs further improve supply chains, aerospace, automotive design, and utilities by optimising monitoring, forecasting, and maintenance.

4.1. DT in personal medicine

DTs are transforming healthcare by creating dynamic virtual replicas of physical entities. These models integrate real-time data, simulations, and predictive analytics to improve medical outcomes. They play a key role in personalised medicine, clinical decision-making, and public health management. This study explores DTs by examining their applications, underlying technologies, benefits, and challenges. By analysing these aspects, we aim to understand their impact on modern healthcare and their potential to improve patient care.

4.1.1. Applications of DT in personal medicine

Digital Twins (DTs) are revolutionising healthcare, advancing both personalised treatment and public health strategies. Kamel Boulos and Zhang highlight DTs' integration with electronic health records (EHRs), genomics, wearables, and environmental sensors to enable real-time simulations for disease prediction and personalised care [110]. Papachristou et al. introduce Digital Human Twins (DHTs),

which replicate individual physiological profiles, proving effective in cardiology, diabetes management, and surgical planning [111].

Sahal et al. expand the concept through Personal Digital Twins (PDTs), integrating biological, mental, physical, and social data to support precision care in areas such as COVID-19 response, cancer therapy, and osteoporosis prevention [112]. Okegbile et al. focus on Human Digital Twins (HDTs) for early diagnosis, tailored treatments, and surgical simulations [113]. Venkatesh et al. further demonstrate HDTs' role in virtual clinical trials and pandemic management [114].

4.1.2. Underlying DT technologies used in personal medicine

Digital Twins (DTs) depend on a strong technological base. This includes artificial intelligence (AI), big data analytics, the Internet of Things (IoT), cloud computing, and virtual reality (VR). Armeni et al. and Cellina et al. highlight the role of AI and big data in improving simulation and prediction accuracy [115,116]. These tools support virtual clinical trials, which reduce the need for human subjects and ease ethical concerns. Gaebel et al. proposed a modular system using natural language processing (NLP) and machine learning (ML) for combining diverse medical data [117]. This enhances clinical decision support systems (CDSS). Papachristou et al. emphasise IoT, cloud computing, and VR for real-time data in Digital Human Twins (DHTs) [111]. Sahal et al. and Okegbile et al. stress the need for blockchain with AI and IoT to manage data securely [112,113]. They also note that ultra-reliable, low-latency communication (URLLC) is vital for real-time sync between physical and digital systems.

4.1.3. Benefits of using DT in personal medicine

Digital Twins (DTs) provide major benefits in clinical decision-making, patient care, and public health. Sun et al. show that DTs improve diagnosis by simulating disease progression and predicting treatment outcomes, enabling timely and effective interventions [118]. This results in better patient outcomes and reduced healthcare costs. Abd Elaziz et al. highlight DTs' ability to support early diagnosis, disease prediction, and treatment optimisation [119]. These functions improve care quality, boost patient involvement, and lower expenses. Papachristou et al. demonstrate DTs' role in surgical planning, helping reduce complications through virtual simulations [111]. DTs also enhance patient engagement by offering personalised health insights. Venkatesh et al. emphasise their use in public health emergencies, such as COVID-19, for efficient risk management and resource planning [114]. Sahal et al. describe Personal Digital Twins (PDTs) as valuable tools for mental and nutritional health monitoring, supporting overall well-being [112].

4.1.4. Challenges involved in the use of DT in personal medicine

Despite their potential, implementing Digital Twins (DTs) in healthcare presents several challenges. Data privacy remains one of the most pressing concerns. Kamel Boulos & Zhang, Papachristou et al., and Sahal et al. highlight ethical issues around patient representation, data ownership, and algorithmic bias [110–112]. Armeni et al. further warn that biases in AI training data can lead to inequitable healthcare delivery [115]. Integration of DTs into existing systems is also complex. Okegbile et al. and Venkatesh et al. point to difficulties in handling heterogeneous data and ensuring interoperability [113,114]. Technical barriers such as high computational demands and the need for robust infrastructure also persist. Meijer et al. emphasise gaps in data standardisation, non-invasive data acquisition, and advanced analytics [120]. Financial burdens add to the complexity, as Cellina et al. note the high costs of developing and maintaining DT systems [116]. Additionally, Venkatesh et al. raise concerns about clinician trust, regulatory uncertainty, and the need for transparent AI systems [114]. Addressing these challenges is key to realising DTs' full potential in transforming healthcare.

4.2. Applications of DT in training and education

DT technology is being widely adopted across diverse fields, including education, training, cybersecurity, and industry. While its primary goal is to improve efficiency, learning, and security, its implementation varies across domains. In education, DTs enhance hands-on learning experiences, while in safety training and cybersecurity, they strengthen preparedness and threat mitigation. Additionally, the

integration of Augmented Reality (AR) and Virtual Reality (VR) further expands DT capabilities, creating immersive and interactive environments. The following study explores the different approaches to DT adoption, highlighting their role in advancing knowledge, security, and operational effectiveness.

4.2.1. Educational applications of Digital Twin

Zhou et al. and Kuhn et al. examine the role of DT technology in education and training, employing different methods and targeting distinct audiences [121,122]. Zhou et al. explored how augmented reality (AR) enhances chemical engineering education by enabling students to visualise and interact with complex chemical processes [121]. By integrating AR with Computational Fluid Dynamics (CFD) simulations, students gain a better understanding of airflow dynamics and the internal structures of lab equipment. Their findings show that AR-driven DT learning improves engagement, reduces safety risks, and boosts technical proficiency. In contrast, Kuhn et al. developed a modular DT training system aimed at non-technical learners in industrial settings [122]. Their approach uses low-cost training kits with microprocessors, sensors, and actuators to teach data analysis and system optimisation via open-source tools like Unity and Node-Red [121,123]. Both studies underscore DT's educational impact [121,122].

4.2.2. Digital Twin for safety and training

Speiser & Teizer and Martínez-Gutiérrez et al. examine DT applications in safety-critical training environments [124,125]. Speiser & Teizer introduce the Digital Twin for Construction Safety (DTCS) framework, which uses real-time construction data to create personalised safety training scenarios that adapt to changing site conditions [124]. This dynamic approach enhances hazard recognition and supports risk-based decision-making in complex construction settings. In contrast, Martínez-Gutiérrez et al. investigate the integration of DT with virtual reality (VR) for industrial operator training [125]. Their study compares VR-based training with traditional equipment and computer-based methods. While equipment-based training yields the highest skill improvement (47%), VR training delivers a strong 38% gain, with lower costs and reduced safety risks. The primary distinction is DT-driven scenario personalisation by Speiser & Teizer versus VR's immersive, cost-effective simulation by Martínez-Gutiérrez et al. [124,125].

4.2.3. Cybersecurity applications in Digital Twin

Unlike other studies emphasising education, workforce training, or safety, Kandasamy et al. focus on the role of Digital Twins (DTs) in cybersecurity research [123]. Their work introduces the EPIC DT—an advanced digital replica of an electric power microgrid developed to simulate cyberattacks, evaluate defence mechanisms, and support the development of intrusion detection systems (IDS). Unlike physical testbeds, which are costly and hard to scale, EPIC DT offers a scalable, cost-effective, and flexible environment for cybersecurity testing. It supports various communication protocols, including MMS, GOOSE, MQTT, and Modbus, enabling comprehensive vulnerability assessments across smart grid components. A major contribution of this study is the implementation of tools such as the Attack Designer (AD) and Attack Launcher (AL), which facilitate the systematic deployment of cyberattacks like man-in-the-middle (MITM) attacks. These tools enhance researchers' ability to examine system resilience under different threat scenarios. This work underscores the unique application of DTs in securing critical infrastructure, setting it apart from studies focused on education and training [123].

4.2.4. Key differences in implementation and technologies among the representative articles

Each study integrates distinct technological innovations to enhance the effectiveness of DT applications in different domains. Zhou et al. combine Augmented Reality (AR) and Computational Fluid Dynamics (CFD) simulations to improve visualisation in educational environments [121]. Kuhn et al. emphasise a modular, hands-on approach by incorporating IoT-based tools to support interactive digital literacy training [122]. In contrast, Speiser & Teizer use real-time construction data and game engine technology (Unity) to dynamically generate safety training environments, tailored to changing site conditions

[124]. Martínez-Gutiérrez et al. leverage Virtual Reality (VR) to offer immersive operator training, enhancing situational awareness while minimising physical risks [125].

Kandasamy et al. take a different path, applying DTs in cybersecurity by simulating cyberattacks on an electric microgrid using specialised tools like the Attack Designer and various communication protocols, including MQTT and Modbus [123]. This study uniquely highlights DT's application in infrastructure protection.

While Speiser & Teizer and Kandasamy et al. focus on automation and real-time adaptation, Zhou et al. and Kuhn et al. provide structured, step-based learning [121–125]. Despite these varied approaches, all studies underscore DT's transformative role-enhancing education, safety, or security and suggest future research may merge these capabilities for more comprehensive applications.

4.3. Applications of DT in design and optimisation

Digital twin technology in design and optimisation leverages AI, IoT, simulation, and cloud computing to create virtual replicas of physical systems. It enables real-time performance analysis, predictive simulations, and iterative design improvements, reduce development time and costs. By integrating data-driven insights, digital twins optimise product lifecycles, structural integrity, and energy efficiency. These applications enhance decision-making, streamline manufacturing, and drive innovation across various industries. The following sections analyse the application of DT in design and optimisation from different perspectives.

4.3.1. Application domains of DT in design and optimisation

The selected references highlight the versatility of DT technology across various industries, each addressing specific operational goals. Bellalouna applies DT for product design optimisation by reducing material usage and improving structural performance through stress simulations of an arbour press [126]. Kalantari et al. introduce the Ph2D system in architecture, merging physical modelling with digital analytics to enhance spatial planning and energy efficiency [127]. In facility management, Asare et al. integrate DT with BIM and IoT for predictive maintenance, improving cost-efficiency and decision-making [128]. Sreedharan et al. utilise DT and Industrial IoT to enhance safety and productivity in remote mining operations [129]. Bellavista et al. focus on adaptive DTs to boost resilience and flexibility in Industry 4.0 digital factories [130]. Sifat et al. propose a DT framework for electric grid management, emphasising predictive maintenance and cybersecurity [131]. Lee et al. prioritise occupant-centric DTs for enhanced comfort and energy optimisation in smart buildings [132].

4.3.2. Innovations of DT in design and optimisation

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4.3.3. Benefits and performance outcomes of DT in design and optimisation

The implementation of DTs in the selected studies yields significant performance benefits. Bellalouna achieves notable reductions in material waste and improved sustainability due to precise simulations and optimised designs [126]. Kalantari et al. enhance creative exploration, spatial visualisation, and interdisciplinary collaboration in architectural design through the Ph2D system [127].

Asare et al. report substantial improvements in facility maintenance efficiency, proactive maintenance strategies, and reduced operational costs, enabled by their predictive DT prototypes [128]. Sreedharan et al. achieve enhanced safety, productivity, and resource optimisation in mining operations, highlighting the transformative impact of DT and IIoT integration [129].

Bellavista et al. demonstrate improved adaptability, autonomous operations, and legacy system integration capabilities, enhancing digital factory operations' reliability and flexibility [130]. Sifat et al. report increased grid stability, optimised resource allocation, and enhanced cybersecurity, enabled by their DT framework [131]. Lee et al. significantly improve occupant comfort, energy efficiency, and empowerment in building management, underlining the advantages of democratised DT systems [132].

4.3.4. Challenges and implementation considerations of DT in design and optimisation

Despite their significant benefits, the selected references also highlight key implementation challenges associated with DT technologies. Bellalouna identifies issues with sensor placement accuracy and data reliability, which are critical for achieving structural optimisation in mechanical systems [126]. Kalantari et al. raise concerns about oversimplifying architectural details in digital translations and recommend enhancing digital functionalities for wider usability in architectural design [127]. Asare et al. point to integration, scalability, data security, and high implementation costs as major barriers in facility management DT platforms [128]. Sreedharan et al. emphasise the difficulties of poor connectivity, extreme environmental conditions, and integrating automation with traditional mining operations [129]. Bellavista et al. discuss challenges in real-time data management and adapting to complex industrial dynamics [130]. Sifat et al. mention the intricacies of modelling complex electric grid subsystems and ensuring cybersecurity [131], while Lee et al. highlight concerns with data privacy, intuitive interfaces, and balancing comfort with energy efficiency in smart buildings [132]. This comparative analysis underscores the vast potential of DTs, while also emphasising the need for interdisciplinary innovation to overcome implementation barriers and enable widespread adoption across sectors.

4.4. DT in infrastructure and urban planning

Digital twin technology in infrastructure and urban planning employs IoT, AI, GIS, and simulation models to construct virtual replicas of cities and infrastructure systems. It enables real-time monitoring, predictive maintenance, and optimised resource management for smart cities and sustainable urban development. Applications include traffic flow optimisation, energy efficiency planning, and disaster response simulations to enhance urban resilience. By integrating real-time data and analytics, digital twins improve decision-making and support the creation of more efficient, liveable, and sustainable urban environments.

4.4.1. Application domains of DT in infrastructure and urban planning

The selected references illustrate diverse applications of DT technology across urban systems and infrastructure. Wan et al. emphasise city-scale Digital Twins (DTs) for smart infrastructure and urban planning, advocating policy-driven strategies and cross-sector data integration. A case study in Cambridge highlights improved coordination in transport, housing, and energy [133]. Abdeen et al. focus on Citizen-Centric Digital Twins (CCDTs) to promote transparency, inclusivity, and public engagement in urban governance [134]. Sharifi et al. apply AI-enhanced DTs for stormwater system performance prediction and operational optimisation, addressing urban environmental challenges [135]. Bellalouna uses DT for product design optimisation, improving material efficiency in an arbour press [126]. Gürdür Broo et al. and Marai et al. explore DTs for smart and road infrastructure, emphasising real-time monitoring and predictive maintenance to enhance resilience [136,137]. Ye et al. highlight human-centred Urban Digital Twins (UDTs) for community resilience and climate adaptation in coastal areas [138]. Together, these studies underscore DT's transformative role in infrastructure and urban planning.

4.4.2. Technological specifications of DT in infrastructure and urban planning

The selected references integrate various technological frameworks into their DT implementations. Wan et al. leverage data science and machine learning, complementing traditional urban theories to

enhance predictive capabilities [133]. Abdeen et al. employ volunteered geographic information (VGI), remote sensors, machine learning algorithms, and Application Programming Interfaces (APIs) for dynamic, interactive management [134]. Sharifi et al. utilise AI, machine learning (ML), deep learning (DL), and IoT sensors to enable accurate stormwater predictions and real-time system adjustments [135].

Bellalouna incorporates IoT platforms, computer-aided design (CAD), and finite element analysis (FEA) for structural optimisation, while Gürdür Broo et al. implement a multi-layered DT framework integrating sensors, data analytics, and visualisation tools [126,136]. Marai et al. rely on IoT devices, edge computing, and advanced object detection algorithms for real-time road infrastructure monitoring [137]. Ye et al. propose integrating big data, AI, and multi-agent interaction models within human-centred UDT frameworks to enhance decision-making and resilience [138].

4.4.3. Benefits and outcomes of DT in infrastructure and urban planning

Implementing DT technology yields numerous benefits across these studies. Wan et al. demonstrate DT's effectiveness in identifying systemic urban risks and facilitating policy experimentation [133]. Abdeen et al. report improved urban infrastructure governance through enhanced citizen participation and real-time responsiveness [134]. Sharifi et al. highlight improved stormwater management, optimised resource allocation, and proactive maintenance through AI-enhanced DTs [135].

Bellalouna achieves significant material reductions and structural efficiency improvements, emphasising sustainability [126]. Gürdür Broo et al. enhance infrastructure resilience, operational efficiency, and predictive maintenance through real-time data analytics [136]. Marai et al. note improved road safety, optimised traffic management, and reduced operational risks, while Ye et al. emphasise enhanced community resilience and improved adaptation planning capabilities for climate risks [137].

4.4.4. Challenges and implementation considerations of DT in infrastructure and urban planning

Despite their advantages, DT implementations face notable challenges across urban infrastructure domains. Wan et al. emphasise the need to align technical development with governance frameworks to ensure effective deployment [133]. Abdeen et al. identify issues with data interoperability, quality of crowd-sourced inputs, and integrating advanced AI models [134]. Sharifi et al. discuss difficulties in data fusion, model accuracy, and the lack of standardisation in AI-driven DTs [135]. Bellalouna highlights challenges in sensor placement, data synchronisation, and achieving real-time monitoring [126]. Gürdür Broo et al. cite barriers such as data quality, complex system integration, and the need for multidisciplinary collaboration [136]. Marai et al. point to cybersecurity threats, high implementation costs, and limited detection accuracy in adverse conditions [137]. Ye et al. stress the integration of socio-environmental data, privacy concerns, and the absence of standard methodologies [138]. Overcoming these barriers is vital for realising DT's full potential in urban planning and infrastructure.

4.5. DT in transportation and logistics

Digital twin technology in transportation and logistics integrates IoT, AI, and simulation models to optimise operations and decision-making. It enables real-time tracking of fleets, predictive maintenance of vehicles, and route optimisation to reduce delays and fuel consumption. In logistics, digital twins enhance warehouse automation, supply chain resilience, and demand forecasting through data-driven simulations. These applications improve efficiency, reduce costs, and enhance sustainability in transportation networks.

4.5.1. Application domains of DT in transportation and logistics

The selected references illustrate the diverse applications of DT technology in logistics and manufacturing, each addressing specific operational challenges and goals. Cuñat Negueroles et al. integrate DT with blockchain to enhance data security, traceability, and operational efficiency in vehicle fleet management and supply chains [139]. Z. Zhang et al. apply DT and blockchain to open production logistics, enabling trusted synchronisation and agile decision-making in dynamic manufacturing settings [140]. L. Zhang et al. explore crowdsourcing logistics by combining DT with a four-party evolutionary

game model to optimise logistics strategies amid e-commerce expansion [141]. Hong et al. focus on synchronising production and delivery across enterprises, using DT to improve coordination between manufacturers and third-party logistics providers (3PLs) [142]. Zhao et al. introduce dynamic spatial-temporal knowledge graphs within DT frameworks to enhance production logistics resource allocation [140]. Coelho et al. utilise DT with simulation-based decision tools to optimise in-house logistics, including warehousing and material handling [143]. Greif et al. propose a lightweight DT model for managing bulk material logistics at construction sites, improving resource efficiency and operational planning [144].

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4.5.3. Benefits and outcomes DT in transportation and logistics

DT implementations across these studies deliver substantial performance improvements. Cuñat Negueroles et al. demonstrate enhanced operational efficiency, reduced delays, and improved decision-making in freight transportation through blockchain-enabled DTs [139]. Z. Zhang et al. report improved reliability, operational cost reductions, and increased adaptability in production logistics synchronisation [145].

L. Zhang et al. achieve system stability and efficiency in crowdsourcing logistics, discouraging dishonest behaviour and improving logistics performance through predictive insights [141]. Hong et al. highlight improved production-delivery coordination, cost efficiency, and system resilience under dynamic conditions, favouring bidirectional collaboration over traditional unidirectional models [142]. Zhao et al. demonstrate optimised travel routes, reduced waiting times, and enhanced operational efficiency in production logistics resource allocation [140].

Coelho et al. validate improved decision-making, bottleneck identification, and resource allocation accuracy in in-house logistics, enhancing operational productivity [143]. Greif et al. show significant cost reductions, improved logistics planning, and enhanced sustainability in construction logistics through DT-based strategies, achieving a 25% decrease in truck-related costs [144].

4.5.4. Implementation considerations of DT in transportation and logistics

Despite their advantages, these DT implementations face a range of technical and operational challenges. Cuñat Negueroles et al. emphasise issues of scalability and data validation in centralised logistics systems, underscoring the value of blockchain for improving transparency and trust [139]. Z. Zhang et al. highlight the need for interoperability and optimisation of dynamic decision-making algorithms in complex logistics environments [140]. L. Zhang et al. point to difficulties in designing effective incentive mechanisms and preventing issues like collusion or false reporting in crowdsourced logistics operations [141]. Hong et al. address challenges in maintaining data privacy, synchronising systems across enterprises, and managing disruptions dynamically [142]. Zhao et al. identify performance bottlenecks in real-time logistics operations and the integration of dynamic disturbances into spatial-

temporal models [140]. Coelho et al. note limitations due to reliance on simulations, emphasising the need for real-time system integration and sustainable practices [143]. Greif et al. recognise difficulties in scaling lightweight DTs for broader low-tech industry use and enhancing predictive analytics [144]. Together, these studies underscore DT's transformative potential while identifying key areas for further innovation and collaboration.

4.6. DT application in process optimisation

The selected references demonstrate the wide-ranging applications of DT technology across multiple industries, each addressing distinct operational challenges through advanced digital modelling. Zhou et al. apply DT integrated with self-adaptive genetic algorithms (SAPGAs) to optimise blast furnace operations in ironmaking, reducing coke consumption while improving efficiency [146]. Bellalouna uses DT for optimising harbour press design, focusing on material efficiency and real-time structural performance analysis [126]. Zheng et al. implement DT in oil and gas production by combining physical models, real-time data, and intelligent algorithms to boost output, lower energy use, and prolong equipment lifespan [1]. Bayer et al. leverage DT and model-based design of experiments (DoE) to enhance yields in bioprocessing applications [147]. Davies et al. focus on predictive maintenance using DTs to monitor degradation and optimise asset servicing schedules [148]. Liu et al. integrate DT into manufacturing process planning, enabling real-time evaluation and reuse of machining strategies [149]. Lim et al. apply DT in dynamic product family design, incorporating context-aware modelling to support flexible industrial operations [150].

4.6.1. Technological specifications of DT in process optimisation

The references exhibit diverse technological integrations within their DT frameworks. Zhou et al. combine SAPGAs with autoregressive moving average (ARMA) models, leveraging cloud-based platforms like Apache Spark for real-time analytics [146]. Bellalouna integrates IoT sensors, computer-aided design (CAD), and finite element analysis (FEA) for structural optimisation and material reduction [126]. Zheng et al. employ multi-disciplinary, multi-scale models combined with IoT, artificial intelligence, and virtual/augmented reality to simulate and optimise oilfield operations [1].

Bayer et al. utilise hybrid modelling techniques, combining mechanistic and data-driven approaches with iterative model validation for bioprocess optimisation [147]. Davies et al. incorporate simulation-based digital twins with mathematical degradation models and MATLAB-based dashboards for maintenance decision-making [148]. Liu et al. implement a Digital Twin-based Process Knowledge Model (DT-PKM), integrating geometric information, process constraints, and real-time equipment data for dynamic process evaluation [149]. Lim et al. propose a three-layer digital twin architecture comprising cyber-physical interaction, data processing, and knowledge computation layers. This architecture integrates IoT sensors, cloud computing, and big data analytics to enable context-aware product design [150].

4.6.2. Benefits and outcomes of DT in process optimisation

DT implementations across these studies demonstrate substantial performance improvements. Zhou et al. achieve a reduction in coke consumption during ironmaking by approximately 14 kg per ton of iron. This improvement enhances both the economic and environmental performance of the process [146]. Bellalouna achieves a 60% reduction in material use for an harbour press, enhancing design efficiency and sustainability [126]. Zheng et al. report increased oilfield efficiency, reduced energy costs, and improved production predictions through digital twin optimisation [1].

Bayer et al. demonstrate efficient bioprocess optimisation, significantly reducing experimental efforts from twenty-seven to nine experiments while maximising yields [147]. Davies et al. effectively predict the remaining useful life (RUL) of engineering components, supporting proactive maintenance and reduced downtime [148]. Liu et al. enhance manufacturing efficiency, tripling process planning speed and increasing process knowledge reuse by over 60% [149]. Lim et al. enhance operational safety and efficiency in smart manufacturing by implementing context-aware asset optimisation. This approach effectively addresses supply chain and operational disruptions [150].

4.6.3. Implementation considerations of DT in process optimisation

Despite their benefits, the selected references highlight several implementation challenges associated with DT technologies. Zhou et al. report occasional deviations under abnormal conditions, emphasising the need for robust anomaly-handling mechanisms in optimisation frameworks [146]. Bellalouna identifies issues with precise sensor placement and data synchronisation, both are essential for accurate DT performance [126]. Zheng et al. cite challenges in model accuracy, data integration, and adapting complex DTs to varied oilfield conditions [1]. Bayer et al. address concerns about model bias and reliability when working in high-dimensional design spaces for bioprocessing applications [147]. Davies et al. note limitations from relying heavily on simulated data and recommend validating DT models with real-world assets [148]. Liu et al. discuss narrow application scopes and the need for reliable data acquisition systems [149]. Lim et al. highlight cyber-physical synchronisation issues and challenges in scaling DTs from lab settings to real-world environments [150]. These studies underscore DT's transformative potential while emphasising the importance of overcoming technical, operational, and contextual barriers for broad adoption.

4.7. DT in manufacturing, supply chain management, and logistics

The following section reviewed studies illustrate the diverse applications of DT technology in manufacturing, supply chain management, and logistics. The integration of artificial intelligence (AI), machine learning (ML), and optimisation techniques has significantly enhanced decision-making, efficiency, and resilience across various sectors.

4.7.1. Use of DT in process optimisation in manufacturing

Shan et al. and Yuan et al. both emphasise process optimisation in manufacturing through DT-driven approach. While Shan et al. focus on optimising pharmaceutical manufacturing, particularly in solvent recovery, Yuan et al. address production efficiency in electrical cable manufacturing [151,152]. Both studies implement advanced optimisation techniques Shan et al. using NSGA-II and CNN-LSTM models, while Yuan et al. employ multi-objective algorithms and a piecewise coupling mechanism [151,152]. The outcomes in both cases showcase notable improvements in efficiency, yield, and defect reduction, highlighting DT's potential to refine manufacturing processes.

4.7.2. Use of DT in production layout and internal supply chain optimisation

Lee et al. and Cimino et al. explore DT applications in production layout and supply chain optimisation, respectively [153,154]. Lee et al. focus on enhancing production flexibility in SMEs through a Simulation-Optimisation with Multi-Criteria Decision-Making (SOMCDM) framework, whereas Cimino et al. develop a modular DT platform to optimise internal supply chains [153]. Both approaches integrate simulation models, but Lee et al. utilise Multivariate Adaptive Regression Splines (MARS) for parameter optimisation, while Cimino et al. adopt object-oriented modelling for ISC management [153,154]. The studies differ in scope, Lee et al. target SMEs struggling with Industry 4.0 adoption, whereas Cimino et al. cater to multi-plant production planning, making their findings more relevant for large-scale manufacturers.

4.7.3. Use of DT in supply chain coordination and resilience

Pan et al. and Singh et al. examine DT's role in improving supply chain coordination and resilience [155,156]. Pan et al. propose a multi-level DT architecture for kitting production logistics, incorporating real-time data integration and a cloud-fog-edge-terminal model to optimise operations dynamically [155]. In contrast, Singh et al. analyse DT's influence on resilient and sustainable manufacturing supply chains, identifying key factors through the Grey Influence Analysis (GINA) methodology [156]. Pan et al. focus on synchronisation and efficiency improvements in manufacturing logistics, but Singh et al. take a broader approach by considering environmental and operational sustainability aspects in MSCs [155,156]. Both studies emphasise DT's ability to enhance coordination, but Singh et al. extend the analysis to resilience and sustainability, making their work more applicable to long-term strategic planning.

4.7.4. Use of DT in predictive modelling and resilient production strategies

Corsini et al. and Perno et al. investigate DT applications in enhancing production resilience and predictive modelling [25,157]. Corsini et al. develop a DT-driven replenishment strategy for semiconductor manufacturing using Artificial Neural Networks (ANN) and Particle Swarm Optimisation (PSO) to address supply chain disruptions [25]. Perno et al. implement a machine learning-based DT framework for catalyst production, evaluating multiple algorithms for optimal process monitoring [157]. While both use predictive analytics, Corsini emphasises inventory and supply chain resilience, whereas Perno focuses on real-time process parameter prediction.

4.7.5. Use of DT in environmental and quality control applications

Rietdorf et al. and Fu et al. integrate DT technology with environmental monitoring and quality control frameworks [158,159]. Rietdorf et al. develop a DT prototype for real-time environmental monitoring in battery manufacturing, incorporating Life Cycle Assessment (LCA) and aligning with regulatory standards [158]. Fu et al. propose a simulation-in-the-loop DT framework for additive manufacturing, using finite element analysis (FEA) and image segmentation to detect structural defects [159]. While both emphasise monitoring and optimisation, Rietdorf targets sustainability, whereas Fu improves defect detection and material efficiency.

4.7.6. Use of DT in Human-Robot collaboration in manufacturing

Park et al. distinguish themselves by integrating DT technology with human-robot collaboration (HRC) through the use of exoskeleton-type robots [160]. Unlike other studies that primarily focus on manufacturing optimisation, supply chain resilience, or sustainability, Park et al. emphasise DT's potential to enhance ergonomic and intuitive interaction between humans and robots. Their system incorporates real-time motion tracking, haptic feedback, and virtual reality (VR) visualisation to create a responsive and immersive collaborative environment. This unique approach highlights how DT can extend beyond traditional applications, offering operational and ergonomic benefits in industrial robotics.

While all the referenced studies investigate DT's role in manufacturing, they diverge in their specific areas of emphasis. Shan et al., Yuan et al., Lee et al., and Cimino et al. focus on process and layout optimisation [40,151,152,154], aiming to improve efficiency and throughput. Others, like Corsini et al., address supply chain resilience by using DT to handle disruptions [25]. Real-time monitoring is explored by Rietdorf et al. and Fu et al., targeting environmental compliance and structural integrity in manufacturing processes [158,159]. Predictive modelling is central to Perno et al.'s study [157], while Park et al. spotlight the integration of DT in human-robot collaboration. Together, these works reflect the broad and evolving potential of DT as a key enabler of Industry 4.0 [160].

4.8. Digital Twin technology in healthcare: Advancements, applications, and challenges

DT technology is transforming healthcare by enabling advancements in personalised medicine, clinical decision-making, disease prevention, and public health monitoring. Originally developed for engineering and manufacturing, DTs now play a crucial role in medicine by improving diagnostics, treatment planning, patient monitoring, and hospital operations [110]. A DT is a virtual replica of a physical entity that continuously updates using real-time data from electronic health records, wearable devices, imaging, and biological markers [111]. The following section explores the applications, technological foundations, challenges, and future directions of DTs in healthcare by analysing findings from multiple studies.

4.8.1. Applications of Digital Twins for personalised Medicine and precision healthcare

Digital Twins (DTs) are revolutionising precision medicine by enabling patient-specific modelling for accurate diagnosis, disease prediction, and treatment planning. These models combine data from genomics, wearables, physiological signals, and AI analytics to simulate outcomes and support clinical decisions [110]. Papachristou et al. present Digital Human Twins, integrating IoT, AI, and cloud platforms for real-time health monitoring and chronic disease management [111]. In orthopaedics and cardiology,

DTs enable biomechanical and cardiovascular simulations, as noted by Sun et al. [118]. In oncology, DTs help personalise cancer therapies by integrating imaging, health records, and multi-omics data [116,117] improving treatment precision and adaptability.

4.8.2. Applications of Digital Twins in ophthalmology and neurology

DTs are also being developed for ophthalmology applications. Iliuță et al. present a DT framework for glaucoma prediction, integrating genetic, demographic, and clinical data to assess patient risk [161]. Their model achieves 84 percent accuracy in detecting early glaucoma signs, offering potential for early intervention and prevention.

Neurological applications are explored by Gazerani, who introduces Intelligent Digital Twins for migraine management [162]. These models analyse biosensor data to predict migraine triggers, optimise treatments, and provide personalised interventions. This demonstrates how specialised DTs improve treatment strategies for complex neurological conditions.

4.8.3. Use of DT in pandemic response and population health monitoring

DTs enhance public health surveillance by enabling real-time tracking of disease outbreaks and risk assessment. Kamel Boulos & Zhang highlight their use in COVID-19 monitoring, demonstrating how DTs helped predict outbreaks and guide public health responses [110]. Similarly, Sahal et al. introduce Personal Digital Twins that integrate biological, mental, and environmental data to provide early health warnings and insights [112].

Okegbile et al. and Venkatesh et al. propose Health Digital Twins, which optimise pandemic response, vaccine distribution, and healthcare resource allocation [114,163]. Their research suggests that DTs can improve crisis management by integrating real-time epidemiological data, patient tracking, and hospital capacity monitoring.

4.8.4. Cardiovascular disease management using Digital Twins

Cardiovascular disease is another area benefiting from DT applications. Coorey et al. explore Health Digital Twins in cardiovascular healthcare, particularly for ischaemic heart disease, heart failure, and aneurysm repair [164]. Their study shows that real-time data synchronisation enhances risk assessments and treatment decisions.

Abd Elaziz et al. extend DT applications to remote cardiac monitoring, patient stratification, and surgical planning [119]. By integrating electronic health records and wearable sensor data, DTs can predict cardiac events, support preoperative simulations, and optimise long-term disease management.

4.8.5. Data integration, standardisation, and computational complexity associated with the use of DT in health care system

A major challenge in DT implementation is data standardisation and integration. Meijer et al. highlight difficulties in merging heterogeneous datasets from electronic health records, wearable sensors, and multi-omics sources [120]. Armeni et al. emphasise the computational complexity involved in real-time data fusion and AI-driven modelling [115].

Gaebel et al. propose modular DT architectures to improve interoperability, while Coorey et al. discuss Cyber-Physical Systems and Closed-Loop Optimisation to enhance data synchronisation [117,164]. These approaches aim to make DT adoption scalable and efficient in clinical practice.

4.8.6. Ethical concerns and data privacy risks of using DT in health care system

DTs raise significant ethical concerns regarding data ownership, privacy, and AI biases. Meijer et al. and Armeni et al. highlight privacy risks, algorithmic biases, and clinician scepticism as barriers to DT adoption [115,120].

To address these concerns, Sahal et al. and Gazerani propose blockchain and federated learning as solutions for enhancing data security and reducing AI biases [112,165]. Regulatory frameworks such as the FDA's Medical Device Development Tools program are evolving to ensure AI-based medical technologies meet safety and ethical standards [114].

4.8.7. Future directions and innovations in Digital Twin healthcare

Future directions in digital twin healthcare focus on integrating AI-driven predictive analytics to forecast diseases and optimise personalised treatment plans. Real-time IoT integration will enhance continuous monitoring, allowing for precision healthcare through seamless data exchange. Blockchain technology is expected to play a crucial role in ensuring secure and interoperable data sharing, protecting patient information while fostering collaboration among healthcare providers. Additionally, advancements in genomic and biomolecular simulations will enable more accurate drug testing and genetic therapy simulations, paving the way for truly personalised medicine.

4.8.8. Expanding Digital Twin applications in telemedicine and virtual clinical trials

Future developments will expand DT applications in telemedicine, remote patient monitoring, and virtual clinical trials. Abd Elaziz et al. predict that AI-driven diagnostics will play a crucial role in digital healthcare ecosystems [119]. Papachristou et al. foresee DTs reducing reliance on human control groups by providing virtual patient models for drug trials [111].

4.8.9. Advancements in AI-driven predictive models

Enhancing AI-driven predictive capabilities is crucial to the success of digital transformation. According to Iliuță et al. and Gazerani, deep learning models have the potential to significantly improve early disease detection and enable more personalised treatment approaches [161,162]. The integration of natural language processing will further enhance data interpretation in clinical decision-making.

4.8.10. Integration with emerging technologies

DTs will increasingly integrate with edge computing, blockchain, and the Internet of Things to enhance real-time processing and cybersecurity. Okegbile et al. emphasise the role of secure cloud architecture, while Sahal et al. propose blockchain-based solutions for data transparency and interoperability [112,113].

Digital Twin technology is revolutionising healthcare by advancing personalised medicine, optimising disease management, and improving public health monitoring. Digital Human Twins, Personal Digital Twins, and Health Digital Twins are transforming precision medicine through real-time simulations, predictive analytics, and AI-driven diagnostics.

Despite these benefits, challenges such as data privacy, standardisation, and computational demands must be addressed. Future research should focus on improving AI-driven predictive models, integrating blockchain for security, and expanding DT applications in telemedicine and virtual trials.

As digital twin technology continues to evolve, its potential to enhance patient outcomes, reduce healthcare costs, and revolutionise medical practice remains vast. Continued innovation in this field will pave the way for smarter, data-driven, and patient-centric healthcare systems.

4.9. DT in autonomous systems

DT technology is revolutionising autonomous systems by creating real-time digital replicas of physical systems, continuously updated through sensors and data analytics. These replicas enable monitoring, predictive maintenance, virtual testing, and performance optimisation for autonomous vehicles, drones, and robots. DTs reduce costs, mitigate risks, and accelerate innovation by allowing engineers to test and improve systems safely and efficiently. They also enhance adaptability and safety in dynamic environments. A recent review highlights the growing relevance of DTs across various domains, comparing their applications by domain, methodology, performance, scalability, and future potential.

4.9.1. Application domains in autonomous systems

DT technology is being effectively applied across multiple autonomous system domains, demonstrating its versatility and impact. In autonomous driving, Shoukat et al. [166] leverage DT to integrate hybrid reality with real-time simulations, enhancing safety and test efficiency. In the field of autonomous marine systems, Hasan et al. [167] utilise DT for fault diagnosis, particularly focusing on propulsion systems, enabling precise detection and minimal operational disruption.

In industrial applications, Khawale et al. [165] implement a DT-based hybrid modelling framework for fault mitigation in autonomous diesel engines. Their approach combines data-driven and physics-based methods to achieve high fault detection accuracy and rapid recovery. Agriculture also benefits from DT innovations. De Bortoli et al. [168], applied a DT-based computer vision algorithm to improve the alignment accuracy of autonomous cultivators during weeding operations.

These application domains highlight the adaptability of DT technology to address unique operational challenges ranging from transportation and energy systems to maritime and agricultural automation. Each case underscores how DTs enhance reliability, safety, and performance in autonomous environments.

4.9.2. DT technologies in autonomous systems

A range of DT technologies has been employed to support autonomous system functionality through real-time data processing, hybrid modelling, and simulation integration. In autonomous driving systems, Shoukat et al. [166] combine real and virtual environments using 3D coordinate mapping, collision detection, and virtual scene registration. This hybrid reality approach allows for synchronised interaction between physical vehicles and their digital counterparts.

Khawale et al. [165] introduce a hybrid engine model that integrates neural networks with physics-based equations, implemented within a hierarchical control architecture. This design enables real-time fault detection and recovery in diesel engines with high computational efficiency. Hasan et al. [167] incorporate an Adaptive Extended Kalman Filter (AEKF) algorithm into their DT framework for autonomous ships. This allows precise estimation of fault parameters using real-time sensor data, enhancing diagnostic accuracy.

In precision agriculture, De Bortoli et al. [168] implement a computer vision-based DT system utilising template matching and colour segmentation within the Lab colour space. This facilitates accurate row detection and cultivator alignment under challenging field conditions.

These technologies highlight the wide range of DT applications across sectors; all centred on creating dynamic digital replicas to enable smart decision-making in autonomous systems.

4.9.3. DT performance metrics in autonomous systems

Performance evaluation is critical to validating the effectiveness of DT technologies in autonomous systems. Various metrics have been used across studies to assess accuracy, responsiveness, and stability under operational conditions.

Shoukat et al. [166] conducted performance tests at multiple sampling intervals (50 ms, 200 ms, and 800 ms) and found 200 ms to be optimal. This rate offered a balanced trade-off between real-time responsiveness and system stability in autonomous driving simulations. Khawale et al. [165] reported a 96% fault detection accuracy in their DT-enabled diesel engine model, coupled with rapid recovery times following detected anomalies. The system maintained stable torque output and exhibited minimal overshoot even during fault scenarios.

Hasan et al. [167] demonstrated high diagnostic precision in maritime environments using real-time DT frameworks. Their system is effectively distinguished between normal and faulty states, enhancing fault detection reliability.

These metrics highlight the impact of DT on fault tolerance, system resilience, and predictive capability. The consistent emphasis on real-time performance and diagnostic accuracy underscores DT's potential as a foundational technology in high-stakes autonomous applications.

4.9.4. DT scalability in autonomous systems

The scalability of DT technology in autonomous systems is a central theme across several domains addressed in the selected studies. In autonomous driving, Shoukat et al. present a DT-based hybrid reality test system that combines real and virtual environments [166]. This framework supports variable sampling rates and real-time feedback, enabling adaptability to evolving vehicle models and complex driving scenarios. In diesel engine applications, Khawale et al. propose a scalable DT-enabled fault mitigation system using a hybrid engine model and hierarchical control strategies [165]. Validated through

both simulation and hardware testing, their system can be extended to various engine types and autonomous platforms, including unmanned marine vessels.

Hasan et al. explore scalability in maritime systems, introducing a DT framework for fault diagnosis in autonomous ships [167]. Their system employs the AEKF algorithm to process real-time sensor data and adapt fault estimation to different operational conditions, validated on the Otter vessel. In agriculture, De Bortoli et al. implement a DT-based vision system for autonomous cultivators, with modular algorithms like template matching and adaptive measurement confidence, demonstrating robust performance across varied crop types and field conditions [168]. These examples underscore DT's scalable potential across diverse autonomous domains.

4.9.5. DT unique contributions in autonomous systems

Each DT implementation across autonomous systems introduces distinct innovations tailored to domain-specific challenges, showcasing the versatility and creative potential of DT technology.

Shoukat et al. [166] uniquely integrate DT with hybrid reality, enabling dynamic testing environments that blend virtual and real-world conditions. Their system enhances simulation fidelity and supports real-time interaction for autonomous driving evaluations. Khawale et al. [165] contribute a hybrid engine modelling framework that fuses neural networks with physics-based formulations, offering high fault detection accuracy and rapid system recovery, which is particularly valuable in safety-critical applications.

Hasan et al. [167] validate their DT system for autonomous ships using both simulation and experimental data, demonstrating strong diagnostic precision and real-time responsiveness. Their integration of an adaptive extended Kalman filter allows for nuanced fault parameter estimation. De Bortoli et al. [168] bring innovation to agricultural automation by developing a DT-based vision algorithm that uses template matching and colour segmentation, achieving robust performance even under variable field conditions.

4.9.6. Future research directions of DT in autonomous systems

Future research in DT technologies for autonomous systems emphasise enhancing predictive capabilities, expanding sensor integration, and improving system adaptability across diverse environments.

Several studies propose strengthening the predictive functionality of DT frameworks. Shoukat et al. [166] and Khawale et al. [165] both highlight the need to refine real-time data integration and forecasting in complex, dynamic environments. Improved predictions would enhance system safety and operational planning, particularly in high-risk or fast-changing scenarios.

Sensor fusion and hardware integration are also key areas for advancement. De Bortoli et al. [168] and Khawale et al. [165] suggest incorporating additional sensors to capture more nuanced environmental or operational data, which could improve system responsiveness and decision accuracy. Enhanced sensor integration is especially relevant in fields such as agriculture and heavy machinery, where external conditions vary significantly.

Scalability remains a central concern. Lin et al. [163] and Khawale et al. [165] all envision broader industrial deployment of their DT models, adapting them to new domains such as marine systems, manufacturing, and energy infrastructure. This will require more modular and interoperable DT architectures that can be efficiently customised to fit different use cases.

Overall, the future of DT in autonomous systems lies in more intelligent, adaptive, and context-aware models that can learn from data, interact with complex environments, and scale across sectors without loss of reliability or performance.

4.10. DT technology applications across various supply chains

DT technology has emerged as a transformative force in modern supply chains, offering unprecedented capabilities to optimise operations and enhance decision-making. Selected articles examine DT applications across diverse supply chains, focusing on key properties such as operational impact, scalability, strategic decision-making, resilience, and sector-specific adaptations. Digital Twins enhance operational

efficiency and streamline processes by combining real-time monitoring, predictive analytics, and modular architecture.

DT frameworks facilitate strategic insights and fairness among stakeholders, ensuring robust supply chain resilience. Sector-specific adaptations further demonstrate DT's versatility, addressing unique challenges in areas like agri-food, manufacturing, and logistics.

4.10.1. Operational impact of DT on supply chains

Singh et al. and Cimino et al. both emphasise DT's role in enhancing visibility and coordination [154,156]. Singh et al. highlight real-time monitoring for better decision-making, while Cimino et al. focus on predictive analytics and 'what-if' analyses that improve key operational metrics like flow time and order tardiness.

Cimino et al. uniquely address DT's modular architecture, making it adaptable for multi-plant environments [154]. This property sets it apart from Singh et al., which broadly recommends exploring DT across different industries but does not detail scalability [156].

4.10.2. Strategic decision-making and fairness

Zhang et al. uniquely focus on strategic decision-making and fairness in DT applications [169]. Unlike other studies that emphasise operational efficiencies, Zhang et al. address the interactions among manufacturers, consumers, and regulators. The study highlights the challenges posed by investment costs, tax incentives, and data transparency, emphasising the need for governmental policies to facilitate DT adoption in ESG contexts.

4.10.3. Resilience

Resilience is another crucial property explored differently by Singh et al. [170]. This study highlights the roles of supply chain resilience (SCR) and supply chain performance (SCP) as key mediators of DT technology's sustainability impact. Using Structural Equation Modelling to demonstrate DT's positive effects on agility, waste reduction, and customer satisfaction.

4.10.4. Sector-Specific adaptations

Sector-specific adaptations are prominently featured in Gallego-García et al. and the refrigerated supply chain research [171,172]. Gallego-García et al. adapt DT technology specifically for agri-food supply chains, focusing on smallholder vulnerabilities and integrating tools like SWOT and Material Flow Analysis. In contrast, refrigerated supply chain research uniquely addresses temperature management and biochemical quality of perishable fruits, highlighting DT's precise applications in product-level logistics.

These studies demonstrate DT technology's diverse properties: operational enhancements, scalability, strategic decision-making and fairness, resilience, and sector-specific adaptations [171,172]. Future research should further investigate these properties across broader contexts and industries.

4.11. DT technology in oil and gas

DT technology has emerged as a transformative force in the oil and gas industry. This review highlights the recent research on DT applications in oil and gas, focusing on technology integration, application scope, real-time capabilities, innovation, and industry impact. Several studies emphasise diverse integration strategies, ranging from cloud and edge computing to SCADA, GIS, IoT, and advanced AI methods. Research spans broad operational areas, such as predictive maintenance and reservoir simulation, to specialised functions like pipeline management and leak detection. Furthermore, real-time monitoring and adaptive learning innovations highlight DT's potential to enhance safety, efficiency, and decision-making. By evaluating these distinct criteria, the review underscores DT's transformative impact and identifies future research directions to further optimise operational performance significantly in diverse environments.

4.11.1. *Technology integration*

Knebel et al. emphasise DT integration with cloud and edge computing, providing scalable computational power and low latency [173]. Similarly, Bo et al. highlight integration with SCADA, GIS, and IoT for centralised pipeline management [174]. Wei et al. integrate 3D modelling and AI methods like Case-Based Reasoning (CBR), distinct from Knebel and Bo's focus on infrastructure integration [175]. Liang et al. and Wen et al. integrate DT with deep learning and hybrid control algorithms, respectively, showing a more specialised technology fusion [176,177].

4.11.2. *Application scope*

Knebel et al. broadly address various oil and gas operations, from predictive maintenance to reservoir simulation [173]. In contrast, Bo et al. narrowly focus on pipeline management [174]. Wei et al. target offshore oil/gas field development, emphasising FEED processes [175]. Liang et al. further narrow their focus to leak detection, while Wen et al. concentrate on natural gas flowmeter calibration, representing the most specialised scope [176,177].

4.11.3. *Real-Time capabilities*

Knebel et al. highlight cloud-edge architectures enabling real-time data analysis and remote monitoring [173]. Similarly, Liang et al. prioritise real-time anomaly detection through continuous adaptive learning [176]. Bo et al. acknowledge real-time monitoring challenges, while Wei et al. and Wen et al. both incorporate real-time data to enhance visualisation and calibration precision, respectively [174,175,177].

4.11.4. *Innovation*

Liang et al. present a significant innovation with adaptive learning in leak detection, outperforming traditional methods [176]. Wen et al. innovate by combining physics-based models with advanced machine learning controls, achieving high calibration accuracy [177]. Wei et al. introduce innovative AI-driven case-based reasoning for rapid FEED design [175]. Knebel et al. and Bo et al. innovate mainly through integration strategies rather than novel methodologies [173,174].

4.11.5. *Industry impact*

Knebel et al. argue DTs enhance overall operational efficiency, safety, and decision-making [173]. Bo et al. stress pipeline safety, cost reduction, and sustainability [174]. Wei et al. highlight efficiency improvements in offshore projects, reducing design time and costs [175]. Liang et al. provide impactful improvements in leak detection accuracy and reliability [176]. Wen et al. significantly impact calibration practices, reducing time and operational costs [177].

These studies highlight DT's diverse benefits in oil and gas based on distinct criteria. Knebel et al. and Bo et al. broadly integrate existing technologies for infrastructure management [173,174]. Wei et al. and Wen et al. focus on specific operational improvements [175,177]. Liang et al. uniquely innovate in real-time detection capabilities [176]. Future research should further investigate these criteria to expand DT's transformative potential in various operational contexts.

4.12. *DT technology in aerospace applications*

DT technology is emerging as a transformative tool in the aerospace industry, driving improvements across monitoring, assembly, machining, and security. This comprehensive analysis examines four significant studies that explore diverse aspects of digital twins. Selvarajan et al. emphasise uncertainty reduction in aerospace monitoring through deep learning, while Aggarwal et al. focus on enhancing data privacy and security using blockchain technologies [178,179]. Jin et al. extend digital twin applications to aerospace assembly by integrating advanced models for reconfigurable tooling [180]. Liu et al. introduce a biomimicry-inspired method to optimise machining processes with real-time monitoring and adaptive control [181]. Together, these studies illustrate the multifaceted potential of digital twins to revolutionise aerospace operations by improving efficiency, precision, and safety in complex environments. This review highlights current trends and future potential.

4.12.1. Application domain and focus of DT in aerospace applications

Selvarajan et al. use digital twins to improve aero engine monitoring, replacing manual inspections with virtual models for higher accuracy under extreme conditions [178]. Aggarwal et al. explore data privacy in cloud-based DT platforms, enhancing cybersecurity for aerospace data exchange [179]. Jin et al. apply DTs to aerospace assembly, monitoring reconfigurable tooling systems to ensure quality and shorten production cycles [180]. Liu et al. integrate biomimicry into DT-based monitoring of aerospace machining, enabling real-time adjustments and adaptive control for improved precision during complex manufacturing processes [85]. Together, these studies highlight diverse DT applications in the aerospace industry.

4.12.2. Methodological approach of DT in aerospace applications

Selvarajan et al. integrate aero transfer functions with deep learning models, specifically employing Convolutional Neural Networks (CNN) and Radial Basis Networks (RBN) [178]. This combination is designed to optimise data representation and reduce uncertainties, ensuring improved performance in high-speed conditions. Aggarwal et al. take a markedly different route by incorporating blockchain technology with elliptic curve cryptography (ECC), fuzzy verifiers, and BAN logic [179]. This integrated approach not only secures data exchange but also minimises computation and communication overhead. Jin et al. develop a comprehensive digital twin system that merges geometric, physical, behavioural, and rule-based models, embedding calibration parameters to address variations and improve monitoring accuracy [180]. Liu et al. propose a novel modelling method inspired by biological systems [181]. By combining geometric, behavioural, and contextual sub-models with an XML-based data integration framework, they enable adaptive real-time updates that capture dynamic changes during the machining process.

4.12.3. Performance and key findings of DT in aerospace applications

The performance outcomes reported by these studies vary according to their specific objectives. Selvarajan et al. report achieving a data transmission efficiency of 91% while reducing uncertainties to 8%, a result that underscores the effectiveness of deep learning in high-demand aerospace environments [178]. Aggarwal et al. demonstrate remarkable efficiency by reducing computation time to 178.297 ms and lowering communication costs to 5184 bits [179]. Their approach also provides strong resistance against various cyber-attacks, highlighting its real-world applicability. Jin et al. show that their digital twin system effectively detects positioning deviations and optimises assembly sequences, thus reducing manual inspections and minimising errors in aerospace assembly [180]. Liu et al. highlight that their biomimicry-based approach outperforms traditional systems by significantly enhancing machining precision and enabling predictive maintenance through real-time data fusion and adaptive control [181].

4.12.4. Security and data privacy of DT in aerospace applications

Security and data privacy emerge as a critical focus, particularly in the work of Aggarwal et al. [179]. Their blockchain-based approach ensures data verifiability and integrity by integrating cryptographic techniques and secure authentication protocols. This emphasis on protecting sensitive aerospace data sets their study apart from the others, which primarily focus on performance enhancements and operational efficiency. While Selvarajan et al., Jin et al. and Liu et al. prioritise reducing uncertainties and optimising monitoring processes, Aggarwal et al. provide an indispensable security layer for digital twin applications in increasingly interconnected aerospace environments [178–181].

4.12.5. Future research directions of DT in aerospace applications

Looking towards the future, each study outlines clear avenues for further development. Selvarajan et al. suggest that integrating artificial intelligence for multi-item detection could further enhance control and predictive capabilities within digital twin systems [178]. Aggarwal et al. propose further incorporation of AI and machine learning to boost big data analytics, thereby expanding the utility of their security framework in real-world scenarios [179]. Jin et al. recommend the adoption of advanced sensors and AI-driven analytics to further refine real-time monitoring and system effectiveness in aerospace

assembly [180]. Finally, Liu et al. aim to improve modelling speed through cloud technologies and to address simulation lag, ensuring that their biomimicry-based approach remains both scalable and robust [181]. Collectively, these future research directions underscore the evolving nature of digital twin technology in aerospace, where performance, security, and adaptability are continuously refined.

4.12.6. Accuracy improvement of DT in aerospace applications

Selvarajan et al. utilise deep learning (CNN, RBN) with aero transfer functions to minimise uncertainties in aerospace systems, achieve a notable 8% uncertainty reduction [178]. Liu et al. also focuses on accuracy through biomimicry-based DT modelling, improving machining precision and real-time predictive accuracy [181]. Jin et al. enhance accuracy in reconfigurable tooling by embedding calibration parameters and rule models, improving detection of positioning deviations [180].

4.12.7. Adaptability of DT in aerospace applications

Liu et al. employ biomimicry-inspired models (geometric, behavioural, contextual) for dynamic and adaptive DT updates, reflecting physical changes during machining in real-time [181]. Jin et al. provide adaptability through reconfigurable tooling systems, supporting multiple product types. Selvarajan et al. use deep learning for adaptive uncertainty management, primarily in predictive maintenance, Aggarwal et al. do not emphasise adaptability [178,179].

Digital twin technologies are revolutionising aerospace applications by enhancing monitoring, precision, and security. Their innovative methodologies drive operational efficiency and robust data management across diverse aerospace processes. Future research will likely yield even greater integration of AI, advanced sensors, and cloud computing to address current limitations and foster further advancements.

4.13. DT technology in the automotive sector

DT technology is transforming the automotive sector by enabling precise virtual models of physical vehicles and systems. It fosters technological innovation through advanced simulations that optimise design, manufacturing, and maintenance. DT allows manufacturers to predict outcomes, reduce development time, and improve performance. Its adaptability spans various applications, including autonomous driving, connected vehicles, and smart manufacturing, making it vital for managing complex systems and integrating emerging technologies. DT also plays a key role in promoting sustainability by supporting efficient resource use, reducing waste, and minimising environmental impact. Additionally, it enhances lifecycle management and improves vehicle safety and reliability. By enabling continuous improvement and innovation, DT helps deliver safer, more efficient, and environmentally responsible automotive solutions. These transformative capabilities position DT as a foundational technology for the industry's future.

The following section reviews recent advancements in DT research within the automotive domain, based on four core criteria: technological innovation, practical impact, adaptability, and sustainability.

4.13.1. Application of DT in energy management in electric vehicles

Ye et al. demonstrate technological innovation by integrating DT with deep reinforcement learning (DRL) for electric vehicles (EVs). This innovation leads to practical impacts, improving energy efficiency by 7.08% and reducing battery degradation by 25.28%. Their approach showcases adaptability through real-time updates based on driving conditions. Sustainability is indirectly supported by extending battery lifespan, thus reducing waste and resource consumption.

4.13.2. Application of DT in riding profile prediction for motorcycles

Smeets et al. exhibit technological innovation by developing a comprehensive DT model incorporating motorcycle mechanics, rider behaviour, and environmental factors [182]. The practical impact is high, as accurate predictions of speed and banking angles enhance safety systems and vehicle design. Their model demonstrates adaptability by integrating various data sources and adjusting to different riding conditions. Sustainability aspects are less emphasised, with the focus primarily on performance and safety.

4.13.3. Circular economy strategies using DT

Mügge et al. focus significantly on sustainability by utilising DT to optimise end-of-life (EoL) decisions, supporting circular economy (CE) practices [183]. Technological innovation lies in creating transparent, product-specific DT models across vehicle lifecycles. Practical impacts include improved resource efficiency and sustainable dismantling. Adaptability is demonstrated by applying the framework across different vehicle components and scenarios, although real-time adaptability is less prominent.

4.13.4. Classic car restoration using DT

Ferreira et al. bring technological innovation by integrating IoT sensors with DT for classic car restoration [184]. Practical impacts include enhanced workshop efficiency, accurate resource management, and authenticity preservation. Adaptability is present through real-time monitoring and predictive analytics tailored to unique restoration needs. Sustainability is indirectly addressed through improved resource management but is not the primary focus.

In technological innovation, Ye et al. and Ferreira et al. stand out for integrating advanced technologies (DRL, IoT sensors) [184,185]. Smeets et al. are innovative by incorporating human factors [182]. Mügge et al. innovate through comprehensive lifecycle modelling [183].

Practical impacts are strong across all studies. Ye et al. and Ferreira et al. significantly enhance operational efficiency [184,185]. Smeets et al. impact vehicle safety and design, while Mügge et al. improve sustainability practices [182,183].

These studies highlight DT's diverse applications and strengths across criteria like innovation, impact, adaptability, and sustainability. Integrating these criteria can develop robust, multi-functional DT systems beneficial for future automotive industry advancements.

4.14. Digital Twin in energy system

Digital twin technology is revolutionising energy systems by creating real-time digital replicas of physical assets. These models enable continuous monitoring, predictive maintenance, and simulation of various operational scenarios across generation, transmission, and distribution. They enhance system efficiency, reliability, and fault detection, allowing for proactive decision-making. Digital twins also support renewable energy integration and advanced grid management, contributing to sustainability and reduced operational costs. By optimising performance and enabling innovation, they help modernise energy infrastructure and ensure resilient, secure operations.

The following section presents a detailed analysis of selected articles on digital twin applications in the energy sector.

4.14.1. Application domains and key focus areas on energy system

The available studies highlight the diverse and impactful applications of DT technology in energy systems. Jamil et al. optimise energy management in smart nano grids by integrating Artificial Intelligence of Things (AIoT) and blockchain, enhancing energy trading efficiency [186]. Spinti et al. apply DTs and Bayesian Decision Theory to biomass energy systems, focusing on operational efficiency and emission reduction at a biomass plant [187]. Deakin et al. develop DT frameworks for electrical distribution networks, improving state estimation, voltage regulation, and microgrid energy trading [188]. Padovano et al. emphasise sustainability, using cognitive DTs to support lifecycle assessment and waste management [189]. Sifat et al. propose a comprehensive DT framework integrating Internet of Everything (IoE) and Industrial IoT (IIoT) for grid stability and cybersecurity [131]. Manfren et al. develop interpretable DTs for building energy management using smart thermostatic radiator valves and heat pumps, reducing residential energy use [190]. Other innovations include Mousavi et al.'s DT and deep learning system for monitoring floating wind turbines [14], Bardeeniz et al.'s transfer learning-enhanced DT for thermal spray dryers [191], and Diz et al.'s edge-computing DT for power converter maintenance [192].

4.14.2. Technological innovation of DT in energy system

Technological innovation is diverse across studies. Jamil et al. utilise AIoT and blockchain, highlighting advanced data security and trading optimisation [186]. Spinti et al. innovate by applying Bayesian decision theory to biomass energy systems, integrate real-time sensor data [187]. Deakin et al. implement real-time power flow algorithms, addressing dynamic distribution system challenges [188]. Padovano et al. introduce a lifecycle assessment-based framework for managing digital waste, while Sifat et al. integrate IoE, IIoT, and machine learning for predictive grid maintenance [131,189]. Manfren et al. develop lean, interpretable regression models for building energy management [190]. Mousavi et al. combine deep learning with simulated data for structural health monitoring [14]. Bardeeniz et al. use LSTM-based transfer learning for spray dryer efficiency, and Diz et al. leverage PSO and GA algorithms for real-time power converter monitoring [192,193].

4.14.3. Practical impact of DT in energy system

All studies demonstrate significant practical impacts. Jamil et al. reduce peak load and increase renewable energy use [186]. Spinti et al. improve boiler efficiency and reduce emissions in biomass systems [187]. Deakin et al. achieve substantial energy curtailment reductions [188]. Padovano et al. effectively identify and mitigate digital waste [189]. Sifat et al. enhance grid stability and operational efficiency [131]. Manfren et al. significantly reduce natural gas consumption in buildings [190]. Mousavi et al. effectively detect structural damage in wind turbines [14]. Bardeeniz et al. considerably reduce energy demand and emissions in spray dryers [191]. Diz et al. efficiently detect component degradation in power converters [192].

4.14.4. Adaptability of DT in energy system

Adaptability is well demonstrated across the references. Jamil et al. showcase real-time adaptability in energy trading and monitoring [186]. Spinti et al. dynamically update operational strategies every five minutes [187]. Deakin et al. include dynamic voltage control and energy trading adaptability [188]. Padovano et al. apply their methodology across various CDT lifecycle stages [189]. Sifat et al. offer a modular, scalable system design [131]. Manfren et al. capture real-time energy performance variations [190]. Mousavi et al. validate their approach under varying environmental conditions [14]. Bardeeniz et al. demonstrate adaptability by transferring learning between simulations and real applications [191]. Diz et al. highlight real-time adaptability through edge computing [192].

4.14.5. Sustainability of DT in energy system

The selected studies demonstrate varying degrees of focus on sustainability. Padovano et al. and Manfren et al. explicitly target sustainability through lifecycle assessment and reduced energy consumption [190,194]. Jamil et al. support sustainability by promoting renewable energy use in smart nano grids [186]. Spinti et al. and Deakin et al. indirectly enhance sustainability through emission reductions and operational efficiency improvements [188,195], while Sifat et al. integrate renewable energy to promote sustainable grid management [131]. Mousavi et al. and Diz et al. contribute by extending infrastructure lifespan and minimising maintenance demands [14,192]. Bardeeniz et al. directly reduce energy use and emissions in industrial processes [191].

Technological innovation across these studies includes deep learning, real-time monitoring, and advanced algorithms. The practical impact is significant, improving efficiency, reliability, and system performance. Adaptability is evident in real-time responsiveness and scalability. While sustainability approaches vary, all studies contribute either directly or indirectly through efficiency gains and reduced environmental impact.

4.14.6. Technological integration and optimisation methods of DT in energy system

Artificial intelligence and machine learning are common across these studies. Jamil et al. and Sifat et al. emphasise blockchain and cybersecurity in energy trading and grid management [131,186]. Optimisation methods differ among the studies. For example, Jamil et al. and Diz et al. use heuristic approaches like Particle Swarm Optimisation [186,192]. Spinti et al. uniquely apply Bayesian methods

[187]. Transfer learning appears only in Bardeeniz et al.'s work [191]. Performance metrics often focus on energy efficiency and sustainability, as noted by Jamil et al., Bardeeniz et al., and Manfren et al. [186,190,191]. In contrast, predictive accuracy and real-time control are emphasised by Spinti et al., Deakin et al., Sifat et al. and Diz et al. [131,188,192].

4.14.7. Unique technological features and challenges

Technologically, the studies differ in their focus and approach. Manfren et al. stress interpretability and actionable insights for building energy management [190]. Mousavi et al. apply digital twin technology for offshore wind turbine structural health monitoring [14]. Padovano et al. highlight lifecycle management and environmental impacts [189]. Other studies place less emphasis on these aspects. Common challenges include data quality and model uncertainty. These issues are noted by Deakin et al., Spinti et al., and Mousavi et al. [188,196].

4.14.8. Future research directions

Future research may focus on integrating advanced machine learning techniques. Researchers are exploring ways to increase scalability and improve automated decision-making capabilities. These findings show that digital twins have broad applicability. They also reveal a varied technological focus. Digital twins can improve operational efficiency, sustainability, and technological advancement across many sectors. This research paves the way for future interdisciplinary collaboration. Further studies will unlock even more potential in digital twin technology.

4.15. DT in marine and offshore Engineering

Digital twin technology plays a significant role in marine and offshore engineering. It creates accurate digital replicas of vessels, offshore platforms, and marine structures. Engineers use these models for real-time monitoring and to simulate various operating conditions. Digital twins help predict maintenance needs and prevent potential failures. They also support design optimisation and enhance safety measures. This technology improves operational efficiency and reduces downtime. It aids decision-making by providing detailed data on structural health and environmental impacts. Digital twins drive innovation and offer promising solutions for the challenges faced in the marine and offshore industry.

The following section explores references on various applications of DT technology across the marine and offshore sectors, each with distinct focuses and methodologies.

4.15.1. Applications of DT in marine renewable energy

Majidi Nezhad et al. discuss the transformative role of digital twins in marine renewable energy [197]. They emphasise operational efficiency, predictive maintenance, and environmental sustainability. Their work follows a holistic approach that integrates IoT, AI, and big data analytics. The aim is to achieve extensive data interoperability. They also target a wide range of applications, such as offshore wind and carbon capture.

4.15.2. DT applications for offshore structural safety

Fang et al. focus on predicting fatigue crack growth in offshore platforms [195]. They use finite element surrogate modelling and dynamic Bayesian networks. Their primary aim is to ensure structural safety and precise maintenance. They rigorously validate their approach with experiments on material specimens. They emphasise real-time updating and accuracy in handling complex and variable loading conditions.

4.15.3. Use DT in floating wind turbine (FWT) management

Liu et al. and Mousavi et al. both focus on Floating Wind Turbines (FWTs) [14]. Their methodologies differ significantly [198]. Liu et al. present a comprehensive DT framework that integrates lifecycle data [198]. Their work emphasises simulation, real-time monitoring, and predictive maintenance. The framework follows a structured, multi-layer approach. They validate their method with real-world installations, offering practical insights into lifecycle management and cost efficiency. In contrast, Mousavi

et al. combine DT with deep learning techniques [14]. They use a Deep Convolution Long Short-Term Memory Neural Network (DCLSTMNN) for advanced structural health monitoring. Their work relies on simulated data and sophisticated signal processing. This method accurately detects structural damage under uncertain loading conditions, reducing the need for physical inspections.

4.15.4. Applications for DT in ecological and coastal management

Pillai et al. stand apart with their focus on ecological and coastal management [194]. They apply digital twin technology to evaluate seagrass effectiveness in mitigating storm surges. Their method integrates numerical models with ecological variables, such as vegetation dynamics. Their validation approach is scenario-based and offers significant insights into climate resilience strategies. This method is distinctly different from the engineering-centric approaches used by other references.

4.15.5. Common challenges for DT in marine and offshore Engineering

Common challenges among the references on use of digital twin applications in Marine and Offshore Engineering include data integration complexities, cybersecurity concerns, and uncertainties in real-world conditions. There is also a need for standardised frameworks. Majidi Nezhad et al. and Liu et al. address these issues within marine energy systems [197,198]. Fang et al. and Mousavi et al. focus on technical challenges related to structural health uncertainties [14,195]. Pillai et al. uniquely address ecological modelling challenges [194]. They work on flexible vegetation dynamics and real-time data assimilation.

4.15.6. Summary and future directions

In summary, each selected reference uniquely contributes to digital twin applications. Majidi Nezhad et al. provide broad best practices [197]. Fang et al. focus on structural safety [195]. Liu et al. offer life-cycle management frameworks [198]. Mousavi et al. enhance structural monitoring through deep learning. Pillai et al. pioneer ecological applications [14,194].

Future research directions include advancing data interoperability and AI analytics. They also call for refining models with real-world data and integrating flexible environmental dynamics. These studies highlight digital twin's diverse potential in marine and offshore applications.

4.16. DT in civil engineering

Digital twin technology is revolutionising civil engineering by creating accurate digital replicas of physical structures. These virtual models enable real-time monitoring, predictive maintenance, and improved decision-making throughout design, construction, and management. Engineers can simulate complex scenarios to assess structural performance and optimise resource use, leading to early detection of issues and proactive maintenance. This technology promotes innovation, sustainability, and efficiency in infrastructure projects, offering benefits like enhanced safety, cost reduction, and improved overall performance. It is shaping a more resilient and future-focused civil engineering industry.

4.16.1. Scope and application

The selected references cover a broad range of applications for DT technology in civil engineering. Pregnolato et al. focus on implementing DTs for existing infrastructure maintenance [199]. Wang et al. address automated post-earthquake structural inspections [196]. Adeagbo et al. explore DTs in advanced rail transit systems [193]. Torzoni et al. discuss predictive maintenance frameworks for civil structures [200]. Sun et al. and Chen et al. concentrate on improving 3D point cloud accuracy for DT modelling [199,201]. Hielscher et al. focus on reinforced concrete bridges [202]. Chiachío et al. present a DT framework for general structural health monitoring (SHM) [203]. Xu et al. integrate DT with Building Information Modelling (BIM) for buildings, and Teng et al. leverage DTs for bridge damage detection using transfer learning [196,204].

4.16.2. Methodological approaches

Selected references adopts distinct methodological approaches. Pregnotato et al. provide a structured five-step workflow integrating finite element modelling [199]. Wang et al. use graphics-based DT frameworks involving UAVs and photorealistic modelling [196]. Adeagbo et al. implement bi-directional communication and adaptive data fusion [193]. Torzoni et al. utilise probabilistic Bayesian networks combined with deep learning [200]. Sun et al. and Chen et al. develop a Cost Volume Pyramid-Based Depth Completion (CVP-DC) deep learning model [205,206]. Hielscher et al. employ neural networks combined with fibre-optic sensor data [202]. Chiachío et al. integrate IoT sensors with Bayesian inference and Petri nets [203]. Xu et al. merge BIM with real-scene 3D models, and Teng et al. apply CNNs trained with DT-generated data and transfer learning [196,204].

4.16.3. DT technologies integrated in civil engineering

The references demonstrate diverse technological integrations. Pregnotato et al. primarily use finite element models and sensor data [199]. Wang et al. integrate UAVs, finite element models, and computer graphics [196]. Adeagbo et al. use sensors, AI, and adaptive data fusion [193]. Torzoni et al. incorporate dynamic Bayesian networks and deep learning [200]. Sun et al. and Chen et al. utilise deep learning and multi-view stereo algorithms [205,206]. Hielscher et al. implement fibre-optic sensors and deep neural networks [202]. Chiachío et al. employ IoT sensors, Bayesian learning, and Petri nets [203]. Xu et al. integrate IoT sensors with BIM and real-scene 3D models [203]. Teng et al. combine CNNs, transfer learning, and Bayesian optimisation [204].

4.16.4. Practical impact of DT in civil engineering

All studies demonstrate significant practical impacts. Pregnotato et al. enhance predictive maintenance for infrastructure [199]. Wang et al. improve safety and efficiency in post-earthquake inspections [196]. Adeagbo et al. enhance rail system monitoring and decision-making [193]. Torzoni et al. track structural health effectively, optimising maintenance [200]. Sun et al. and Chen et al. improve 3D model completeness, aiding accurate structural analysis. Hielscher et al. achieve precise bridge monitoring [201,202]. Chiachío et al. autonomously assess structural conditions, improving maintenance schedules [203]. Xu et al. offer comprehensive real-time building monitoring [203]. Teng et al. enhance structural damage detection accuracy [204].

4.16.5. Adaptability of DT in civil engineering

Adaptability is consistently strong across studies. Pregnotato et al. propose a flexible workflow adaptable to various infrastructures [199]. Wang et al. dynamically adapt UAV inspection paths [196]. Adeagbo et al. emphasise modularity and adaptive updating rates. Torzoni et al. utilise continuous Bayesian updates [200]. Sun et al. and Chen et al. adapt their models to different structures [205,206]. Hielscher et al. use adaptable neural network models based on sensor data [202]. Chiachío et al. demonstrate adaptability in workflow management [203]. Xu et al. adapt by integrating BIM and real-scene models. Teng et al. adapt CNN models from simulated to real conditions using transfer learning [204].

4.16.6. Sustainability of DT in civil engineering

Sustainability is mostly addressed indirectly. Pregnotato et al. suggest improved efficiency leading to sustainable practices [199]. Wang et al. support sustainability by efficient disaster response [196]. Adeagbo et al. aim for long-term asset management, indirectly promoting sustainability [193]. Torzoni et al. focus explicitly on reducing lifecycle costs and improving safety [200]. Sun et al. and Chen et al. support sustainability by improving long-term monitoring accuracy [205,206]. Hielscher et al. and Chiachío et al. indirectly enhance sustainability through extended structural lifespans [202,203]. Xu et al. and Teng et al. contribute to sustainability by facilitating predictive maintenance, reducing waste [204,207].

4.16.7. Validation and practical implementation of DT in civil engineering

Wang et al., Sun et al., Chen et al., and Teng et al. provide substantial real-world validation through practical case studies, enhancing their credibility [196,201, 204,205]. Hielscher et al. also provide experimental validation with high precision results [202]. In contrast, Pregnolato et al., Torzoni et al., and Chiachío et al. primarily validate their frameworks through synthetic or laboratory-scale models [199,200,203]. Xu et al. effectively demonstrate validation through the integration of real BIM and real-scene data [207].

4.16.8. Strengths and challenges of DT in civil engineering

The selected references reveal several strengths. They show methodological clarity and enable rapid, safer inspections [196,205]. Other studies demonstrate real-time monitoring capabilities and support predictive maintenance [200,207]. Accuracy in 3D modelling is highlighted by Sun et al. and Chen et al. [201,205]. Hielscher et al. emphasise precision with sensor data [202]. Chiachío et al. focus on autonomous decision-making [203]. Teng et al. show improved accuracy through transfer learning [204]. Common challenges include standardisation, data validation, and integration complexity. Other challenges are data security, high costs, sensor management issues, and extensive training requirements.

Collectively selected references highlight the transformative potential of DT technology in civil engineering, particularly emphasising structural health monitoring, predictive maintenance, and response optimisation. Differences in scope, methodological approaches, integrated technologies, and validation strategies underscore the diverse and evolving nature of DT applications, demonstrating significant promises but also considerable challenges in implementation.

4.17. DT in industrial automation and Robotics

Digital twins are rapidly transforming industrial automation and robotics by providing dynamic, real-time simulations of physical assets and processes. By integrating sensor data with advanced algorithms, digital twins allow predictive maintenance, optimisation, and performance analysis, reducing downtime and enhancing operational efficiency. In robotics, they enable virtual testing of robotic systems, improving design and functionality before physical deployment. This technology bridges the gap between digital models and the real world, fostering innovation in automation strategies and manufacturing processes. As digital twin applications evolve, they are poised to drive substantial improvements in reliability, safety, and productivity across industries. Embracing these innovations consistently. The following section outlines innovative technologies and applications from multiple perspectives.

4.17.1. Application domains of DT in industrial automation and Robotics

Based on the available references on uses of DT in robotics and automation, several key application domains have emerged prominently. Z. Zhang et al. and Dallel et al. centre their research on enhancing Human-Robot Collaboration (HRC), specifically within assembly tasks, utilising sophisticated algorithms and virtual reality, respectively [208,209]. Baratta et al. adopt a broader analytical approach, reviewing DT applications across various manufacturing processes such as assembly, material handling, and welding, without introducing novel experimental setups [210].

4.17.2. Methodological innovations of DT in industrial automation and Robotics

Distinct methodological innovations appear among the references. Z. Zhang et al. use advanced algorithms such as Occlusion Robust Mesh Recovery (ORMR) and Enhanced Spatiotemporal Graph Convolutional Networks (EST-GCN) [206]. Their approach improves human modelling and action recognition accuracy. In contrast, X. Zhang et al. focuses on knowledge management and structured data representation [211]. They integrate Knowledge Graphs (KG) and Function Blocks (FB) for robotic machining. Lee et al. incorporates Deep Reinforcement Learning (DRL) within DT frameworks [212]. Their method supports adaptive task allocation in unpredictable construction environments and shows significant efficiency gains.

4.17.3. User-Centric innovations of DT in industrial automation and Robotics

Niermann et al. and Li et al. prioritise ease of use and user engagement [143,213]. They focus on user-centric innovation. Niermann et al. propose a visual programming interface integrated with DT. Their design reduces the cognitive load for operators. It makes complex automation tasks accessible to non-experts. Li et al. combine Augmented Reality (AR) with DT [213]. Their approach enhances intuitive teleoperation and improves coordination in multi-robot systems. Their work demonstrates flexibility and user-friendly interaction.

4.17.4. Validation and outcomes of DT in industrial automation and Robotics

In terms of validation and outcomes, several studies highlight substantial performance improvements and safety enhancements. Z. Zhang et al., Dallel et al., and Lu et al. validate their DT systems in realistic scenarios, showcasing marked advancements in operational efficiency, accuracy, and safety [208,209]. Hu distinguishes their research through remarkable improvements in vision-guided robotic grasping, achieving near-perfect success rates via mutual information-driven DT, significantly outperforming conventional methods [209]. X. Zhang et al. and Lee et al. emphasise adaptive process control and optimisation, achieving notable reductions in process steps and time, reflecting DT's robust predictive and adaptive capabilities [141,212].

Overall, each selected representative reference underscores DT's potential across different domains, they diverge significantly in their methodological approaches, user engagement strategies, and validation outcomes, collectively illustrating DT's multifaceted capabilities in improving industrial processes and human-robot interactions.

4.18. Application of DT in various supply chain domains

The available six references explore the role of DT technology across various supply chain domains, each focusing on different applications, methodologies, key findings, technological integrations, sustainability aspects, and future research directions. Although all the studies recognise the transformative potential of digital transformation (DT), they each apply it within different industrial contexts, utilising distinct analytical methods to tackle specific challenges. Together, these studies underscore the increasing importance of DT in strengthening supply chain resilience, improving efficiency, promoting sustainability, and enhancing decision-making processes.

4.18.1. Major uses of DT in supply chain domains

The application of DT varies across different supply chain environments. Singh et al. focus on manufacturing supply chains, with the former emphasising resilience and sustainability, and the latter exploring how DT influences resilience and supply chain performance to drive sustainability outcomes [156,170]. Cimino et al. investigates the optimisation of internal supply chain management in manufacturing through a simulation-based DT platform [210]. Zhang et al. uniquely examines ESG decision-making in semiconductor supply chains, considering the role of fairness concerns and regulatory policies in DT-enabled environments [169]. Gallego-García et al. extends DT applications beyond manufacturing to the agri-food sector, focusing on smallholder farmers who often face challenges related to resource allocation and sustainability [171]. Defraeye et al. addresses the cold chain logistics of perishable food products, particularly mangoes, leveraging DT for improved temperature management and quality preservation [172]. These diverse applications demonstrate the versatility of DT in tackling different supply chain complexities across industries.

4.18.2. Technologies enabling the use of Digital Twin in supply chain domains

Each study employs a distinct methodology to analyse DT's impact. Singh et al. utilises Grey Influence Analysis (GINA) to identify 17 key resilience and sustainability factors in manufacturing supply chains, highlighting visibility and coordination as critical elements. Cimino et al. adopts a simulation-based DT platform, designed with object-oriented structures, to conduct 'what-if' analyses and optimise internal supply chain configurations [210]. Zhang et al. applies an evolutionary game model to study strategic interactions among manufacturers, consumers, and government regulatory agencies in the

semiconductor industry [169]. Singh et al. uses Partial Least Squares Structural Equation Modelling (PLS-SEM) to quantify DT's influence on supply chain resilience and performance based on data from 203 industry professionals [170]. Gallego-García et al. combines system dynamics modelling with simulation techniques to develop the 'Farmer Manager 4.0' framework, helping farmers predict disruptions and optimise processes [171]. Defraeye et al. integrates a biophysical fruit simulator with DT modelling to assess thermal behaviour, enabling real-time tracking of temperature fluctuations in refrigerated supply chains [172]. These methodologies illustrate the diverse analytical approaches used to evaluate DT's role in enhancing supply chains.

4.18.3. Key findings in the use of Digital Twin in supply chain domains

The studies highlight significant insights into DT's impact on supply chains. Singh et al. finds that DT improves supply chain visibility, coordination, demand forecasting, and energy efficiency, thereby enhancing resilience and sustainability [170]. Cimino et al. demonstrates that DT-driven simulations significantly reduce scheduling inefficiencies, improve production flow, minimising tardiness, and optimising resource allocation in internal supply chains [156,210]. Zhang et al. emphasises the role of fairness concerns in ESG decision-making, revealing that high investment demands and low regulatory tax rates hinder manufacturers' motivation to adopt ESG measures [169]. Singh et al. establishes that DT strengthens supply chain resilience and performance, indirectly contributing to sustainability through better waste reduction, quality control, and customer satisfaction [170]. Gallego-García et al. shows that DT enables smallholder farmers to improve decision-making, mitigate risks, and enhance operational efficiency through real-time monitoring and predictive analytics [171]. Defraeye et al. highlights that DT enhances cold chain logistics by identifying temperature fluctuations that impact fruit quality, reduce food waste and post-harvest losses [172]. These findings underscore DT's role in driving efficiency, resilience, and sustainability across various supply chain sectors.

4.18.4. Technologies integration in the use of Digital Twin in supply chain domains

DT integration varies significantly across studies based on industry requirements. Singh et al. highlights how DT improves real-time tracking, decision-making, and risk management in manufacturing supply chains [156]. Cimino et al. leverages DT with predictive analytics and simulation-based architectures to optimise supply chain performance dynamically [210]. Zhang et al. focuses on DT's role in ESG transparency and governance, particularly its influence on regulatory compliance and consumer perception [169]. Singh et al. explores DT's ability to enhance real-time monitoring, coordination, and flexibility in manufacturing operations [170]. Gallego-García et al. integrates DT with emerging Industry 4.0 technologies, such as IoT, big data analytics, and cloud computing, to strengthen agri-food supply chain sustainability [171]. Defraeye et al. utilises DT in combination with real-time environmental monitoring tools to track perishable goods' temperature-sensitive attributes [172]. These integrations demonstrate how DT can be customised to different supply chain needs, offering real-time insights, predictive capabilities, and automated decision-making tools.

4.18.5. Driving sustainability through Digital Twin technology in supply chain operations

Sustainability is a common theme across all studies but with different focal points. Singh et al. emphasises DT's role in sustainable manufacturing, particularly in improving energy efficiency and reducing supply-demand mismatches [156]. Cimino et al. focuses on optimising resource utilisation and enhancing agility in industrial supply chains [210]. Zhang et al. explores DT's influence on ESG decision-making, highlighting how regulatory policies and fairness concerns affect sustainability outcomes [169]. Singh et al. links DT adoption to supply chain sustainability, showing how improved resilience and performance contribute to waste reduction and long-term environmental benefits [170]. Gallego-García et al. prioritises sustainability in smallholder farming by enabling better risk management, resource allocation, and supply chain coordination [171]. Defraeye et al. demonstrates DT's potential in reducing food waste and improving refrigeration efficiency in perishable goods logistics [172]. These studies collectively show how DT plays a crucial role in advancing sustainability across diverse supply chain environments.

4.18.6. Future research directions of Digital Twin technology in supply chain operations

Each study outlines future research directions to further advance DT applications. Singh et al. advocate for expanding DT use across industries to enhance dynamic decision-making and support long-term sustainability strategies [156]. Cimino et al. recommend integrating advanced optimisation algorithms and testing DT scalability in diverse industrial environments [210]. Zhang et al. emphasise the need for stronger regulatory frameworks and fairness mechanisms to promote transparent Environmental, Social, and Governance (ESG) adoption in DT-enabled supply chains [169]. Singh et al. also suggest conducting geographically and industry-specific research to better understand DT's impact on supply chain resilience and performance [170]. Gallego-García et al. propose extending their DT-based sustainability framework to additional agricultural sectors and incorporating life cycle assessment tools [171]. Defraeye et al. highlight the need for real-time data integration and broader application of DTs to other perishable goods, accounting for variables such as moisture loss and chilling injuries [172].

Collectively, these studies underscore DT's transformative role in supply chain management, enhancing efficiency, resilience, and sustainability. As the technology matures, continued research and cross-sector adoption will be vital to unlocking its full strategic potential.

5. Future research scope

The future of DT technology presents numerous research opportunities to enhance its implementation and impact. Despite growing adoption across sectors, key challenges remain unresolved. One of the most critical is the lack of standardisation, which hampers data exchange and system interoperability. As industries like healthcare, manufacturing, energy, and logistics expand DT use, establishing universal data formats and communication protocols becomes essential. Future research should focus on creating best practices, governance models, and ethical guidelines-especially in sensitive domains like healthcare and defence-through collaboration among regulatory bodies.

Artificial intelligence (AI) is central to DT advancement. While current systems leverage machine learning (ML) for predictions, emerging techniques in deep learning and reinforcement learning can improve real-time decision-making. Hybrid models that blend physics-based simulations with data-driven approaches will enhance accuracy. AI-powered DTs capable of adapting to new conditions-so-called self-learning DTs have the potential to transform healthcare, manufacturing, and other complex systems. Generative AI can also enable more responsive, high-fidelity simulations.

Real-time processing is another key area for improvement. Applications such as autonomous vehicles and industrial automation demand low-latency data handling. While cloud computing supports large-scale DTs, edge computing can reduce latency and increase responsiveness. Future research should aim to optimise edge architectures for real-time scenarios and minimise energy consumption. In smart cities, edge-enabled DTs can enhance traffic management, emergency response, and infrastructure maintenance. A hybrid cloud-edge approach will help scale DTs while preserving performance.

Security and data integrity are major concerns. As DTs handle sensitive and high-volume data, the risk of cyberattacks increases. Blockchain offers a promising path to secure, tamper-proof data exchange. However, its integration into DT systems remains underexplored. Future work should develop scalable blockchain frameworks that support high-speed transactions and smart contracts to automate DT operations. Robust cybersecurity mechanisms will be essential to prevent breaches and ensure data privacy.

Expanding DT applications into emerging fields offers exciting potential. In quantum computing, DTs could simulate quantum states to advance material science. In biotechnology, they may model genetic interactions and accelerate drug discovery. Space exploration could benefit from DTs by enabling real-time spacecraft monitoring and mission planning. Research in these frontier areas could open new technological frontiers.

Human-DT interaction is also a growing area of interest. Augmented reality (AR) and virtual reality (VR) can enhance visualisation, training, and decision-making. In healthcare, 3D DTs can assist surgeons with preoperative planning. In manufacturing, AR-driven DTs can provide diagnostics and guidance. Future studies should explore neural interfaces, haptic feedback, and intelligent assistants to improve user engagement and accessibility.

DTs can also support environmental sustainability. They can help optimise resource use, reduce waste, and track carbon emissions. Research should prioritise the development of energy-efficient DT architectures and integration with smart grids to enhance energy management. In urban planning, DTs can simulate environmental impacts and optimise transportation networks. They also have applications in biodiversity monitoring and climate change mitigation.

Finally, the ethical, legal, and social implications (ELSI) of DTs must be addressed. Privacy, accountability, and fairness are crucial—particularly as DTs influence decisions in healthcare and governance. Ethical AI frameworks, data ownership regulations, and workforce transition strategies are needed to ensure responsible adoption.

Advancing DT technology requires cross-sector collaboration and focused research in AI, edge computing, cybersecurity, and ethics. These efforts will drive DTs towards becoming more intelligent, secure, and sustainable tools for digital transformation.

6. Conclusion

The rapid advancement and widespread adoption of DT technology across various industries highlight its growing significance and impact. As demonstrated in this systematic review, DTs have evolved from a conceptual innovation into a practical, multifunctional framework with real-world applications spanning healthcare, infrastructure, manufacturing, energy, and transportation. By replicating physical systems in real time, DTs enable continuous monitoring, predictive analytics, and adaptive control, thereby enhancing efficiency, resilience, and sustainability.

A key advantage of DTs lies in predictive maintenance and risk management. Through real-time data integration, machine learning, and dynamic feedback loops, organisations can anticipate failures, minimise downtime, and extend asset life leading to cost savings and enhanced safety, particularly in sectors like aerospace, oil and gas, and marine engineering.

DT implementation relies heavily on enabling technologies such as artificial intelligence (AI), the Internet of Things (IoT), edge computing, and secure communication protocols. These components support decentralised decision-making and seamless physical-digital integration. Moreover, blockchain and cybersecurity frameworks are increasingly being integrated to ensure data integrity and system trustworthiness—addressing critical concerns about security and reliability.

Beyond operations, DTs are reshaping healthcare through patient-specific models for diagnostics and treatment planning, and enhancing training and education via immersive, risk-free simulations. In urban planning, DTs facilitate sustainable infrastructure design, efficient resource use, and improved public services.

To fully harness DT potential, future research must address challenges in scalability, interoperability, standardisation, and cybersecurity. Expanding DT capabilities for interconnected systems and developing ethical and governance frameworks will be critical as these technologies grow in autonomy and complexity.

In conclusion, DT technology is a cornerstone of the ongoing digital transformation, offering powerful tools for operational agility, innovation, and sustainable development. Interdisciplinary collaboration and investment in foundational technologies will be essential to ensure DTs continue to evolve in a rapidly changing world.

Author contributions

All authors contributed significantly to the development of this work. Sk Hasan conceptualised the study, designed the methodology, visualisations and led the writing of the manuscript. Colin Crawford contributed to data collection, analysis, and interpretation. All authors reviewed and approved the final version of the manuscript.

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